Intraday Momentum

A follow-up to the momentum deep dive: which of its ideas survive inside one trading session, and how they fail

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Abstract

Within a trading day, prices mostly revert rather than trend. The one robust exception is late-day continuation. A market that has moved strongly tends to move a little further in its last half hour. Traders who must trade in the direction of the move cause it. The effect is small. It has weakened over time. It is profitable only in the cheapest markets to trade.

Prices absorb news in seconds within a day, so intraday momentum is a different bet from monthly momentum. Short-term moves in individual stocks tend to reverse over minutes to an hour. Dealers charge for providing immediacy, and prices recover once the order pressure ends. Continuation needs traders who must buy after rises and sell after falls. Leveraged ETFs rebalance near the close. Short-gamma options dealers re-hedge. Stop-loss orders trigger as prices cross levels. This trading concentrates late in the day. Across more than 60 futures markets and 45 years, a 1% move in a stock-index future before the last half hour predicted about 0.04% more in the same direction. A partial reversal followed over the next day or two. The first half hour also predicts the last. This better-known result mostly fails out of sample and depends heavily on the 2008 crisis.

Trading costs decide the outcome. A shorter holding period shrinks the expected move but not the cost per trade. A 30-minute hold needs a forecast about 16 times sharper than a one-month hold to break even. Only the large stock-index futures and index ETFs meet this requirement. No documented intraday momentum in individual stocks survives costs. The capacity of the best-documented effect is a few million dollars a year. Large funds have not removed the effect because this capacity is too small for them.

Most intraday findings come from measurement errors. Bid–ask bounce looks like reversal. Stale index components look like morning momentum. A one-bar timestamp error looks like predictive skill. The number of ways to slice a day makes the best backtest look good by chance. About one day in 20 looks like a clean trend day at random. The chapter recommends three things. Start from a named group of forced traders, not a chart pattern. Trade only the cheapest index futures. Test every result on simulated random data first. For slower strategies, the most reliable use of this research is execution timing.

How to read this chapter. This chapter is a sequel. It assumes Momentum in Financial Markets, called *the main document* throughout, and re-derives nothing from it. The chapter sorts the main document's ideas into three groups when the holding period shrinks from months to minutes or hours. Some still hold. Some change sign. Some stop meaning anything. Sections 1–3 build the model. They define intraday momentum, show which mechanisms survive the shorter time scale, and describe the autocorrelation pattern inside a day. Section 4 surveys the evidence effect by effect. Section 5 is the main section of the chapter. It takes the main document's concepts one at a time and says how each translates. Sections 6–7 cover engineering: costs, capacity, execution and testing. Sections 8–9 cover practice and synthesis.

If you have time for one section, read §5. If you are about to backtest an intraday signal, read §7 first. Appendix A explains from first principles the market-microstructure and statistical concepts the text uses in passing. These include auctions, price impact, bid–ask bounce, out-of-sample R^2 and others. Its index says where each concept is first used.

What this chapter takes as known. Other chapters in the collection develop the following concepts. This chapter references them and does not repeat them.

Concept	Where it is developed
Incomplete adjustment, and the four explanation families for momentum	Sections 1.2–1.3 of the main document
The variance ratio as the unifying diagnostic	Sections 1.2 and 4.5.1 of the main document
The master form of momentum signals, the taxonomy and its equivalences	Section 5 of the main document
Volatility normalisation and signal transforms	Section 4.3 of the main document
Information coefficient, hit rate, multiple testing, the deflated Sharpe ratio	Section 7 of the main document
Order-flow long memory, the propagator model, order-flow imbalance, order-book learning	Section 2.6 of the main document

Concept	Where it is developed
Trend P&L as a weighted sum of autocovariances; the variance-ratio identity	Sections 5.2–5.4 of Trend-Following
Metaorders, risk-management flows and leveraged-ETF rebalancing as sources of trend	Section 2.4 of Trend-Following
Dealer gamma, the feedback multiplier, hedging-induced intraday autocorrelation, zero-day options	Sections 5–7 of Dealer Hedging

Epistemic tags. As in the main document:

- **[Fact]** — replicated across independent datasets or implementations, with broad agreement.
- **[Contested]** — documented, but researchers disagree about its magnitude, robustness or cause.
- **[Hypothesis]** — a proposed mechanism, not decisively tested.
- **[Practice]** — practitioner convention. It may be right, but the evidence is private or absent.

Notation. Trading days (sessions) are indexed $d = 1, \dots, D$ and bars within a day $i = 1, \dots, M$. A 6.5-hour US equity session has $M = 13$ half-hours, 78 five-minute bars or 390 one-minute bars. $p_{d,i}$ is the log price at the end of bar i on day d, $p_{d,0}$ the opening price, and $p_{d-1,M}$ the previous close. The bar return is $r_{d,i} = p_{d,i} - p_{d,i-1}$ and the overnight return is $r_d^{\text{on}} = p_{d,0} - p_{d-1,M}$. Windowed returns use the main document's notation: $r_{d,a:b} = p_{d,b} - p_{d,a}$.

The literature names five windows of the day, and this chapter uses the same names. The windows are **ON** (overnight, previous close to open), **FH** (first half hour) and **MID** (the middle, from the end of FH to an hour before the close). The last two are **SLH** (second-to-last half hour) and **LH** (last half hour). Two composites matter. The first is ONFH $=$ ON $+$ FH. The second is **ROD**, the *rest of the day*, $=$ ON $+$ FH $+$ MID $+$ SLH. ROD covers everything from the previous close to thirty minutes before today's close. So r_d^{LH} is the last half-hour return and r_d^{ROD} is the return from yesterday's close to 30 minutes before today's.

Other recurring symbols:

- σ_d is the close-to-close volatility of day d. s_i^2 is the share of a typical day's variance that falls in bar i, called the intraday profile. Bar 0 stands for the overnight period, and $\sum_{i=0}^{M} s_i^2 = 1$, so that $\text{Var}(r_{d,i} \mid \sigma_d) \approx \sigma_d^2 s_i^2$. σ_D is a typical daily volatility used in rough calculations. It is a representative constant, not the volatility of any particular day.
- $\gamma(i,k) = \text{Cov}(r_{d,i},\, r_{d,i-k})$, computed across days, is the **periodic autocovariance**. It depends on the time of day i as well as the lag k. $\rho(i,k)$ is the corresponding correlation.
- $\text{VR}_d = \left(\sum_{i=1}^{M} r_{d,i}\right)^2 / \sum_{i=1}^{M} r_{d,i}^2$, summed over the day's intraday bars only (excluding the overnight bar), is the **per-day variance ratio** (§3.4).
- Q is an amount of order flow in dollars. In Kyle's linear impact, it means signed net buying. In the square-root law, it means the unsigned size of one trader's order (§2.5, §6.5). λ is the price impact per dollar of net flow, which is Kyle's lambda in section 2.6 of the main document. λ_W is λ for a window W later in the day (§1.3). g_j is the **flow convexity** of participant class j, the dollars it must trade in the direction of the market per unit of return (§2.5). $G = \sum_j g_j$.
- h is the holding period of a position, σ_h the return volatility over it, c the round-trip trading cost as a fraction of notional, and IC the information coefficient. V is daily dollar volume. Y is the prefactor of the square-root impact law (§6.5).
- ℓ is the leverage multiple of a leveraged ETF ($\ell = 2, 3, -1, -2, -3$).

As in the main document, two symbols have more than one use. β is a predictive-regression slope in §4 and a factor loading in §5.6. α appears only as a significance level. Each use is local.

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1. The question: does momentum survive inside the day?

1.1 The wrong intuition: momentum on a faster clock

Most people start from the assumption of scale invariance. If prices trend over months because they adjust slowly to news, they should trend over minutes for the same reason, only faster. Under this assumption, a trader takes the twelve-month signal, shrinks it to sixty minutes and runs it on one-minute bars. Every charting package builds in this assumption through its time-frame selector. It offers the same moving average, the same breakout rule and the same RSI on any bar size.

Three facts contradict this assumption. Each is a theme of this chapter.

1. **Information is not slow inside the day.** The main document's first mechanism, slow diffusion of information, operates over weeks. Inside the day, public information is priced in seconds. The S&P 500 ETF (SPY) and the E-mini futures contract (ES) respond to macroeconomic surprises within five milliseconds (Chordia, Green & Kottimukkalur, 2018). Prices fully incorporate analysts' views broadcast live on CNBC within a minute when the views are positive and within about fifteen minutes when they are negative (Busse & Green, 2002). Even in 1984, with news arriving on a teleprinter, the returns to trading on earnings announcements disappeared within five to ten minutes (Patell & Wolfson, 1984). Order flow stays predictable for longer, but price does not. For actively traded NYSE stocks, Chordia, Roll & Subrahmanyam (2005) find that past returns predict nothing even over five-minute intervals. Order imbalances predicted returns for up to thirty minutes in 1996, ten minutes in 1999 and five minutes in 2002.
2. **The day is not a stationary process.** Volatility, volume and spreads follow a deterministic U shape across the session. The flows that move prices arrive on a schedule: the open, the close, benchmark fixes, option expiries and scheduled announcements. The main document's unifying diagnostic, the variance ratio of a stationary return series, has no single value inside a day. Its value depends on *when* in the day it is measured.
3. **Costs do not shrink with the horizon.** The gross edge of a trade scales with the information coefficient times the volatility over the holding period. That volatility shrinks with the square root of the horizon. The cost of a round trip does not shrink at all. So the IC needed to break even rises as one over the square root of the holding period. It is about sixteen times higher for a thirty-minute trade than for a one-month trade at the same cost per trade (§6.2).

The same indicator on minute bars is therefore a different bet. It is usually a bet *on* a negative autocorrelation (§3.3), made at a cost that is an order of magnitude larger relative to the edge. Less survives the translation than the naive assumption implies. The survivor is **conditional, scheduled continuation**. Index futures and index ETFs show it most. It concentrates in the last half hour of the session. It grows stronger on volatile days and when hedgers hold positions that amplify moves.

1.2 What "intraday momentum" names: five different objects

The phrase names at least five different things. The literature is hard to read until a reader separates them. They differ in what predicts what, at what horizon and for whom.

#	Object	What predicts what	Where it is documented
1	**Market intraday momentum**	The day's return so far predicts the last half hour	Index futures and ETFs; also bonds, commodities, currencies, crypto
2	**Same-time-of-day continuation**	A stock's return in one half hour predicts the same half hour on later days	The US stock cross-section
3	**Microstructure momentum**	Order flow predicts the next seconds to minutes	Order books everywhere
4	**Breakout and trend-day trading**	A move out of an early-session range predicts continuation to the close	Mostly practitioner research

#	Object	What predicts what	Where it is documented
5	**The day–night split**	Where the main document's twelve-month momentum earns its return within the 24-hour cycle	The US stock cross-section

The same five, by mechanism and by who can trade them:

#	Horizon	Proposed mechanism	For a trader without co-location
1	30 minutes, once a day	Flows that depend on the day's move and execute near the close (§2.5)	Positive, small, uneven over time (§4.1–4.2)
2	Days to weeks, at a fixed time	Repetitive institutional trading (§4.3)	An execution-timing tool, not a strategy
3	Milliseconds to minutes	Order splitting and queue dynamics	Flow is predictable and price barely is. Only the fastest firms trade it
4	Hours	A mix of object 1, stop cascades and attention	[Contested]: strong claims, weak independent evidence (§4.8)
5	Overnight versus intraday	Clienteles trading at different times of day (§4.4)	Momentum stocks have *underperformed* during the session

Traders usually mean only objects 1 and 4 by the phrase. Object 2 is a seasonality. Object 3 is the flow prediction that section 2.6 of the main document already covers. Object 5 is a fact about which part of the day earns the main document's momentum. A sixth row would describe the default behind all five: **short-horizon reversal**. An individual security's returns reverse inside the day when nothing scheduled is happening (§2.3).

1.3 The three ideas: flow, clock and cost

Three ideas underlie the rest of this chapter. Each is the intraday counterpart of one leg of the main document's framework.

(I) Flow, not news — the mechanism. Inside the day, *inventory and pre-committed flow* adjust slowly. Information does not. Some participants trade by rules fixed in advance, as functions of the price and the clock. They include dealers hedging short option positions, leveraged ETFs restoring their leverage, stop-loss orders, portfolio insurers, benchmark trackers and funds that rebalance on a schedule. Many of them must trade *in the direction* of the day's move, and many do so at predictable times. Liquidity providers absorb whatever arrives in between. The price must compensate them for holding it, so the default is reversal. The following equation states this compactly:

$$\mathbb{E}\big[\, r_{d,W} \,\big|\, \mathcal{F}_{d,\tau} \big] \;\approx\; \lambda_W \, G \, r_{d,\,\text{prev close:}\tau}, \qquad G = \sum_j g_j .$$

The expected return in a window W later in the day equals the price impact per dollar in that window, λ_W, times a net dollar flow. The day's move so far obliges participants to execute that flow in the window. Each participant class j contributes its **flow convexity** g_j, the dollars it must trade in the direction of the market per unit of return. Positive-feedback traders (short-gamma hedgers, leveraged ETFs, stops) have $g_j > 0$. Negative-feedback traders (long-gamma hedgers, profit-takers) have $g_j < 0$. When G is near zero, liquidity provision leaves reversal as the only pattern. Two corrections apply. Traders who *anticipate* the flow buy before W and move part of the effect earlier. The flow's impact also partly reverts once the flow stops (§2.4, §4.2). Every documented intraday continuation effect estimates $\lambda_W G$ for some window. Every documented decay reflects G shrinking or traders anticipating the flow.

(II) The clock — the autocorrelation pattern. Volatility is periodic and flows arrive on a schedule. So a surface, $\rho(i, k)$, replaces the main document's autocorrelation function ρ_k. It is the correlation between the return in bar i and the return k bars earlier, measured across days. The expected P&L of any linear intraday rule is the inner product of its weights with this surface (§3.2). Momentum appears in a few cells of the surface, chiefly the last half hour against the rest of the day. The

cells near the diagonal are negative. Normalisation must use the time-of-day volatility profile. The variance ratio must be measured per window and per day (§3.4).

(III) The cost wall — the economics. Expected edge per trade is proportional to IC $\cdot\ \sigma_h$, and σ_h grows as $\sqrt{h}$. The cost per trade does not shrink. For a sign strategy, the break-even information coefficient is

$$\text{IC}^* \ = \ \frac{c}{\sigma_h}\sqrt{\frac{\pi}{2}},$$

§6.2 derives this result. Intraday effects survive only where c/σ_h is tiny, as in index futures and the most liquid ETFs, or where the IC is unusually high. The IC is unusually high where the flows behind the effect are mechanical rather than informational. Capacity is also small: at the size that maximises profit, a trader keeps a third of the gross edge (§6.5).

The table maps the three legs onto the main document's framework (section 9.1 of the main document).

Leg	Main document	Inside the day
Mechanism	Incomplete adjustment to information	Incomplete adjustment to *flow*. Information is priced in seconds, while pre-committed, price-contingent flow and dealers' inventory adjust over hours
Signature	The variance-ratio profile VR(q)	The periodic autocorrelation surface $\rho(i,k)$, and the variance ratio per window and per day
Estimation	One master form: a normalised, weighted sum of past returns	Anchored, time-of-day kernels normalised by the intraday profile. The cost wall decides which kernels are worth estimating

1.4 The intraday time-scale map

Section 1.5 of the main document maps the sign of return predictability across horizons from seconds to years. It devotes two rows to the intraday range. This section expands those two rows.

Horizon	Dominant sign in prices	Mechanism	Evidence
Milliseconds to seconds	Flow persistent, price close to a martingale; cross-market lead–lag	Order splitting, queue depletion, latency arbitrage	Budish, Cramton & Shim (2015); Chordia, Green & Kottimukkalur (2018)
Seconds to minutes	Reversal in transaction prices	Bid–ask bounce; inventory; metaorder footprints	Roll (1984); Chordia, Roll & Subrahmanyam (2005)
Minutes to an hour	Reversal for single stocks, close to zero for indices	Temporary liquidity imbalances that resolve within an hour	Heston, Korajczyk & Sadka (2010); Herberger, Horn & Oehler (2020)
Rest of day to last half hour	Continuation, at the index level	Price-contingent flows concentrated near the close	Gao, Han, Li & Zhou (2018); Baltussen, Da, Lammers & Martens (2021)
Last half hour to next 1–3 days	Partial reversal	The price pressure of the close reverts	Baltussen, Da, Lammers & Martens (2021)
Close to next open	Positive drift in US equity futures until about 2020; reversal of end-of-day imbalances	Dealer inventory; risk premia	Boyarchenko, Larsen & Whelan (2023)
Same half hour on later days	Continuation in the cross-section	Repetitive institutional trading	Heston, Korajczyk & Sadka (2010)

The pattern is the one that the flow mechanism of §1.3 predicts. Unconditionally and at short lags, the sign is negative. The positive cells occur at scheduled times, when flows that depend on the day's move have to execute.

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§1 Key takeaways

1. **Momentum is not scale-invariant.** The main document's primary mechanism, slow diffusion of information, is complete within minutes for public news in liquid markets. The twelve-month signal shrunk to sixty minutes is a different bet, usually on a negative autocorrelation.
2. **"Intraday momentum" names five different objects.** Traders usually mean only market intraday momentum (the day's move predicting the last half hour) and breakout trading. The other three are a seasonality, a flow prediction and a fact about where daily momentum accrues.
3. **The mechanism is flow, not news.** Participants whose trades are fixed functions of price and clock create continuation in the window where they execute. They are hedgers, leveraged ETFs, stops and rebalancers. Liquidity providers create reversal everywhere else. Expected continuation is roughly price impact times net flow convexity times the day's move.
4. **The autocorrelation pattern is a surface, not a function.** Predictability depends on the time of day as well as the lag. The positive cells occur at scheduled times. The cells near the diagonal are negative.
5. **The economics is a cost wall.** The IC needed to break even rises as one over the square root of the holding period. It is about sixteen times higher at thirty minutes than at a month. Only the cheapest instruments can meet it.
6. **Reversal is the intraday default.** Any intraday momentum claim must first explain why it beats the reversal that liquidity provision produces at every short lag.

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2. Mechanisms: which explanations survive the shorter clock

Section 1.3 of the main document gives four families of explanation for momentum. They are slow information diffusion, behavioural bias, order flow and risk. This section asks what each predicts at intraday horizons and what the evidence says. Two more families matter at intraday horizons, because they operate on exactly the intraday scale. **Liquidity provision** produces reversal. **Price-contingent flow** produces continuation. The scorecard in §2.7 summarises the results.

2.1 Slow information diffusion: mostly absent inside the day

What it would predict. If news reached traders gradually within the session, prices would drift after announcements for minutes to hours, and a signal that followed the first move would profit.

What the evidence says. For public, hard information in liquid markets, adjustment is complete within seconds to minutes, and has been for decades.

- In 1979–1981 data, Patell & Wolfson (1984) find the initial reaction to earnings and dividend announcements in the first pair of price changes after the release. The returns to simple trading rules disappear within five to ten minutes. Disturbances to variance and serial correlation persist for several hours and into the next day.
- Busse & Green (2002) study analysts' reports broadcast live on CNBC. Prices respond within seconds. Prices fully incorporate positive reports within one minute and negative reports over about fifteen minutes. Traders who act within fifteen seconds earn small but significant profits on positive midday reports.
- In foreign exchange, Andersen, Bollerslev, Diebold & Vega (2003) find that macroeconomic surprises produce *jumps* in the conditional mean at the announcement, with bad news moving prices more than good news.
- Chordia, Green & Kottimukkalur (2018) find that SPY and ES respond to macroeconomic surprises within five milliseconds. The profits from trading quickly are about $19,000 per event in SPY and $50,000 in ES. Order flow has become *less* informative over time, because prices now respond to news directly rather than through trading.

The exceptions. Negative news is slower than positive news in Busse and Green's data. Information that arrives overnight enters prices through the open. Much of Patell and Wolfson's residual effect occurs there. "Late-informed trading" is one of the two explanations Gao, Han, Li & Zhou (2018) offer for market intraday momentum. In this idea, some investors process the morning's information and act on it near the close. Baltussen, Da, Lammers & Martens (2021) argue against it with a test. S&P 500 futures trade actively for fifteen minutes after the cash market closes, so informed traders could act then too. Yet the predictability stops at 4:00 p.m., when index options and leveraged ETFs settle.

Verdict. [Fact] Public, hard information in liquid markets is priced in seconds to minutes. Slow diffusion cannot drive continuation over thirty minutes to hours in index futures. The one exception is the weak form of late-informed trading, which is **[Contested]**.

2.2 Behavioural channels: pressure that reverses within minutes

What it would predict. Loss aversion, overconfidence and limited attention distort individual traders' decisions within the day. If those distortions moved prices persistently, they would create intraday trends.

What the evidence says. They move prices briefly, and the move reverses.

- Coval & Shumway (2005) study proprietary traders in Treasury bond futures on the Chicago Board of Trade in 1998. Traders with morning losses were about 16% more likely to take above-average risk in the afternoon than traders with morning gains. They were willing to buy at higher prices and sell at lower ones. The prices they set reverted within ten minutes, faster than prices set by other traders.
- Barber & Odean (2008) show that individual investors are net buyers of attention-grabbing stocks. These are stocks in the news, with unusual volume or with extreme one-day returns. Berkman, Koch, Tuttle & Zhang (2012) find the intraday pattern. Attention-driven buying by individuals at the open produces high overnight returns, followed by reversal during the trading day. The reversal is concentrated in high-attention stocks.
- Sentiment extracted from social media shows one intraday predictive pattern. Renault (2017) studies StockTwits messages from 2012 to 2016. The change in investor sentiment over the first half hour predicts the SPY return over the last half hour, after controlling for past returns. Novice traders drive the result.

Verdict. [Fact] Behavioural pressure inside the day mostly appears as transient price moves that reverse within minutes to a day. Behaviour can produce continuation only if it keeps generating flow in the same direction through the session. The attention buying behind "stocks in play" (§4.8) is the candidate. That link is a **[Hypothesis]**.

2.3 Liquidity provision and inventory: the source of reversal

The main document treats liquidity provision as the reason one-month returns reverse (Jegadeesh, 1990; Lehmann, 1990). Inside the day it is the dominant force. It is also the baseline any momentum claim has to beat.

The mechanism. Grossman & Miller (1988) model market makers as sellers of immediacy. A trader who must trade now pushes the price away from its fundamental value. The market maker absorbs the order into inventory. The price concession pays for the risk of holding it until an offsetting trader arrives. The price then moves back. Any demand shock that carries no information therefore produces a temporary move. The move reverts over the market maker's inventory horizon, which is negative autocorrelation at that horizon.

The evidence.

- Using NYSE intermediary data, Hendershott & Menkveld (2014) estimate that inventory-driven price pressure averages 0.49%, with a half-life of under a day.
- Heston, Korajczyk & Sadka (2010) attribute the short-term reversal in the US cross-section of half-hour returns to temporary liquidity imbalances lasting less than an hour, and to bid–ask bounce.
- Nagel (2012) shows that the returns to short-term reversal strategies are returns to liquidity provision. They are highest when the VIX is high, and the cost of bearing inventory risk is high with it.
- For German blue-chip stocks, Herberger, Horn & Oehler (2020) test sixteen intraday strategies on five-minute returns. Momentum formation and holding periods run from 15 to 60 minutes, and reversal periods from 60 to 300 minutes. They find no momentum and strong reversal. The reversal is statistically significant but too small to survive a retail investor's costs.
- The pattern also appears at scheduled points. Around the London 4 p.m. fix, Evans (2018) finds that rate changes before and after the fix are negatively correlated across 21 currency pairs in 2000–2013. This differs from other times of day. At the equity close, auction price deviations revert quickly and almost completely (Bogousslavsky & Muravyev, 2023).

Verdict. [Fact] Reversal is the default intraday sign for individual securities. The main document tells the trader to skip the most recent month to avoid a known opposite-signed effect. Intraday, that instruction becomes: skip the last bar or bars and work with midquotes (§5.1).

2.4 Metaorders: continuation that others see first

Section 2.6 of the main document develops the microstructure background. Institutions split large orders (*metaorders*) into many child orders. The signs of those trades have long memory. The square-root law governs how impact grows with size. The propagator model explains why prices stay close to a martingale although the flow is predictable. The intraday question is who can see a metaorder while it is running.

A metaorder executed over several hours pushes the price steadily while it runs. The price then partly reverts. This is continuation during execution. It accrues to whoever detects the metaorder first, and competition to do so is intense.

- van Kervel & Menkveld (2019) find that high-frequency traders initially trade against large institutional orders. For the most informed and longest-lasting orders, they then switch to trading in the same direction, known as *back-running*.
- Hirschey (2021) finds that high-frequency traders' aggressive purchases and sales lead those of other investors. The high-frequency traders anticipate buying and selling pressure from patterns in past trades and orders. In doing so, they raise other investors' costs.
- Chordia, Roll & Subrahmanyam (2005) find that the window in which order imbalances predict NYSE returns shrank from thirty minutes in 1996 to ten in 1999 and five in 2002. The imbalances themselves stayed strongly autocorrelated. The flow stayed predictable, but the price did not.

The SEC's 2010 concept release on equity market structure names "order anticipation" and "momentum ignition" as strategies some investors regard as abusive (Securities and Exchange Commission, 2010). The regulator's decision to name them shows how contested the topic is.

Verdict. [Fact] for the flow: metaorders produce intraday continuation while they execute. **[Contested]** for whether a trader using bars can profit from it. A momentum rule on five-minute bars is a slow, noisy metaorder detector that competes with fast, precise ones. My expectation is that it detects each metaorder last.

2.5 Price-contingent flows: the source of continuation

This family makes the master relation of §1.3 concrete. Each participant below trades by a rule fixed in advance, as a function of the price and the clock. Many trade at predictable times of day.

Leveraged and inverse ETFs. A fund with net asset value A and leverage ℓ holds exposure ℓA. After an underlying return R, its exposure becomes $\ell A(1+R)$ and its net asset value becomes $A(1+\ell R)$. Restoring the target exposure $\ell A(1+\ell R)$ then requires a trade of

$$\Delta X \;=\; \ell A(1+\ell R) - \ell A(1+R) \;=\; A\,\ell(\ell-1)\,R.$$

Because $\ell(\ell-1) > 0$ for every $\ell > 1$ *and* every $\ell < 0$, every leveraged and every inverse fund buys after the market rises and sells after it falls. Its flow convexity is $g = A\,\ell(\ell-1)$. Per dollar of assets, $\ell = 2$ gives $2A$, $\ell = 3$ gives $6A$, $\ell = -1$ gives $2A$, $\ell = -2$ gives $6A$, and $\ell = -3$ gives $12A$. The inverse funds are the most convex. These funds promise a multiple of the *daily* return, so they rebalance near the close. As of February 2009, Cheng and Madhavan

estimated leveraged-ETF rebalancing at 16.8% of market-on-close volume on a day the market moved 1%. They estimated 50.2% on a 5% day. Baltussen, Da, Lammers & Martens (2021) report these estimates (see also Cheng & Madhavan (2009)). Market-on-close orders execute at the closing auction's price (A.3). Shum, Hejazi, Haryanto & Rodier (2016) relate the rebalancing to end-of-day volatility. Ivanov & Lenkey (2018) show that investors' capital flows into and out of these funds offset much of the rebalancing, and in the limit all of it. They find the net effect on late-day returns and volatility economically insignificant in 2006–2014. **[Contested]**.

Dealers who are short gamma. An options dealer hedges *delta*, the amount of the underlying that cancels the sensitivity of its options to small price moves. *Gamma*, Γ, is the rate at which that sensitivity changes as the price moves (A.8). A dealer who hedges delta trades $-\Gamma\,\Delta S$ shares for a price change ΔS, or $-\Gamma S^2 R$ dollars for a return R. When dealers are net short gamma ($\Gamma < 0$), they buy after rises and sell after falls, with $g = |\Gamma| S^2$. When they are long gamma, the sign reverses and they dampen moves. Sections 5–7 of Dealer Hedging give the derivation, the evidence and the zero-day-option complication, so this chapter does not repeat them. The intraday point is timing. A dealer can re-hedge at any time. It has three reasons to finish before the close: overnight risk, capital requirements computed from end-of-day deltas, and better liquidity near the close (Baltussen, Da, Lammers & Martens, 2021).

Stop-loss and take-profit orders. Osler (2003) uses the order book of a large foreign-exchange dealer to show that take-profit orders cluster at round numbers, while stop-loss orders cluster just beyond them. Take-profits are negative feedback, so trends tend to reverse at round numbers. Stop-losses are positive feedback, so trends accelerate once prices cross a round number. Osler (2005) documents the resulting *price cascades*. Exchange rates move unusually fast when they reach levels where stop-losses cluster. Almost 10% of stop and take-profit orders sit at rates ending in 00. The response to stop-losses is larger and longer-lasting than the response to take-profits. Here the flow convexity is zero until prices reach a level. At that level it becomes a sudden large flow. This is the microstructural basis of support, resistance and breakout rules. It also explains why breakouts often overshoot and then fail.

Portfolio insurance and volatility targeting. Rules that cut exposure after losses or when volatility rises sell into declines. Their flow is positive feedback. It usually executes daily near the close. Section 5.10 of Dealer Hedging and section 2.4 of Trend-Following give the mechanics. **[Practice]** for the timing.

Benchmark hedges at a fix. Foreign investors who hedge the currency risk of an equity portfolio must resize the hedge when equity prices move. Many do it at month-end, at the London 4 p.m. fix. Melvin & Prins (2015) find that, in 2004–2013, a market's relative equity appreciation predicts depreciation of its currency before the month-end fix. A partial reversal follows the next day. This is the cleanest example of the master relation outside equities. A flow proportional to a price, executed at a known time, moves the price in the window and then reverts. Ito & Yamada (2017) find related effects at the Tokyo fix, where customer orders are predictably biased towards buying foreign currency.

Rebalancing on a schedule. Bogousslavsky (2016) models investors who rebalance only at particular times. Their trading makes returns autocorrelated at the lags between rebalancing times. This produces intraday seasonality and, when some rebalance early in the day and others late, intraday momentum. The flow follows a schedule and is not price-contingent, but its size depends on earlier shocks. Gao and co-authors cite it as their second explanation.

The closing auction. Much of this flow executes at the closing auction. Bogousslavsky & Muravyev (2023) document that the closing auction's share of daily volume rose from 3.1% in 2010 to 7.5% in 2018, driven by indexing and ETFs. Closing prices usually match the pre-close bid or ask. Price impact at the close is lower than during continuous trading. Deviations of the auction price revert quickly and almost completely.

The table lists the flow convexities. By the master relation, the net sum over the rows at that time of day determines the last half hour's expected return.

Participant	Flow convexity g_j	When it trades	Sign	Evidence
Leveraged and inverse ETFs	$A\,\ell(\ell-1)$	Last half hour, at the close	+	Cheng & Madhavan; Baltussen et al.; offset by capital flows (Ivanov & Lenkey)
Dealers short gamma	$\lvert\Gamma\rvert S^2$	Continuously, completed by the close	+	Baltussen et al.; Dealer Hedging §7
Dealers long gamma	$-\Gamma S^2$	As above	−	Ni et al. (2021); Barbon & Buraschi (2020)
Stop-loss orders	A burst when a level is crossed	When levels are hit	+	Osler (2005)
Take-profit orders	A burst at round numbers	When levels are hit	−	Osler (2003)
Portfolio insurance, volatility targeting	Depends on the rule	Often daily, near the close	+	Trend-Following §2.4
Currency hedgers of equity portfolios	Hedge ratio × equity value	Month-end fix	+ (currency against equity move)	Melvin & Prins (2015)
Liquidity providers	Absorb, then lay off	Continuously	− (with a lag)	Grossman & Miller; Hendershott & Menkveld

The section references in the Evidence column are to the other documents in this collection.

An order-of-magnitude check on size. Suppose leveraged-ETF rebalancing on a 1% day is about half a percent of the day's volume. Cheng and Madhavan's 16.8% of closing volume implies roughly this size when closing volume is a few percent of the day's, as it was around 2010. The square-root law, $I \approx Y\sigma_D\sqrt{Q/V}$ with $Y \approx 1$, then puts the impact of that flow at about $1\% \times \sqrt{0.005} \approx 7$ basis points. Baltussen, Da, Lammers & Martens (2021) estimate a pooled slope of 0.042 for equity index futures. A 1% move before the last half hour predicts about four basis points more in the same direction. Two independent sources give the same order of magnitude. This check shows consistency and is not a measurement. The square-root law describes a single metaorder. Aggregate flow across many traders has closer to linear impact for small imbalances (Patzelt & Bouchaud, 2018). Ivanov and Lenkey's capital flows offset part of the effect. **[Hypothesis]**.

G is not constant. It changes sign when dealers move between long and short gamma. It changes size with leveraged-ETF assets and investors' capital flows. It changes timing as zero-day options are hedged within the day rather than at the close, and as the closing auction grows. Intraday momentum is a property of the *current composition of flow*. So it can decay or flip, and a backtest over forty years is a backtest over several different markets (§7.5).

2.6 Risk premia and the day–night split

The fourth family, risk, predicts *where* in the 24-hour cycle returns accrue. It does not predict continuation within the session. The evidence is directly relevant to anyone trading the main document's momentum.

- **US equities have earned their premium overnight.** Cliff, Cooper & Gulen (2008) and Kelly & Clark (2011) find that, for US equities, returns during trading hours have been close to zero or negative on average. Returns from close to open have been positive.
- **Much of it in one hour.** Boyarchenko, Larsen & Whelan (2023) locate the overnight return in ES futures during the opening hours of European markets. They link it to order imbalances at the previous US close. This is consistent with dealers earning a payment for carrying inventory overnight. Robust overnight reversals follow sell-offs, and modest ones follow rallies. In a July 2026 update, the same authors report that the 2:00–3:00 a.m. window earned about 3.7% a year from 1998 to 2020. This is more than 60% of the contract's 5.9% annualised close-to-close return. The window has averaged close to zero since 2021. The authors explain this by a fall in the dispersion of end-of-day order imbalances from 6.5% to 2.9%, as algorithms slice flow more finely (Boyarchenko, Larsen & Whelan, 2026).
- **Market risk is priced at night.** Hendershott, Livdan & Rösch (2020) find stock returns positively related to beta overnight and negatively during the trading day, in the US and internationally.
- **Momentum earns its return overnight.** Lou, Polk & Skouras (2019) decompose fourteen trading strategies into overnight and intraday components. Momentum and reversal strategies earn their profits entirely overnight. Most other strategies earn theirs entirely intraday. Typically the other component has the opposite sign, which reflects opposing flows from

clienteles who trade at different times of day. Bogousslavsky (2021) adds that size and illiquidity premia accrue in the last thirty minutes of trading. Anomalies such as profitability and idiosyncratic volatility accrue during the day and lose overnight, consistent with mispricing at the open. Akbas, Boehmer, Jiang & Koch (2022) find that stocks earn higher future returns when positive overnight returns frequently precede negative daytime reversals.

Verdict. **[Fact]** for the day–night decomposition in US data, which rests on several independent papers over several decades. **[Contested]** for the mechanism. The split rests on opening prices, which are noisier than closing prices, so measurement error is the first objection. The pattern persists for years and across strategies, which makes pure noise an unlikely explanation. For this chapter the implication is that the main document's twelve-month momentum is, in US stocks, an overnight phenomenon. A trader who buys momentum winners in the morning and sells them at the close holds them through the half of the day in which they have historically *underperformed*.

2.7 The scorecard

Mechanism	Family in the main document	Intraday prediction	Horizon	Status
Slow diffusion of public news	Slow diffusion	Continuation after news	Seconds to 15 minutes	**[Fact]** that it is too fast for anyone but the fastest
Late-informed trading	Slow diffusion	Morning move predicts last half hour	Hours	**[Contested]**
Behavioural pressure	Behavioural bias	Transient pressure, then reversal	Minutes	**[Fact]** for reversal
Liquidity provision	Order flow	Reversal	Minutes to an hour; overnight	**[Fact]**
Metaorders	Order flow	Continuation while executing	Minutes to days	**[Fact]** for flow; **[Contested]** for slow traders
Price-contingent flows	Order flow	Continuation into the window where flows execute, reversal after	Last half hour; fixes	**[Fact]** that it exists; **[Contested]** shares
Risk premia	Risk	Returns accrue overnight	The 24-hour cycle	**[Contested]** mechanism

Inside the day, the main document's four families behave differently. Slow diffusion of public news no longer operates. Behaviour survives mostly as reversal. Risk changes character and becomes a statement about day versus night. Order flow dominates and splits into a reversal half and a continuation half.

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§2 Key takeaways

1. **Slow diffusion does not drive intraday momentum.** Public information in liquid markets is priced in milliseconds to minutes. Even in 1984, the returns to trading on earnings news disappeared within ten minutes.
2. **Behavioural pressure reverses.** Loss-averse traders move prices for minutes, and attention buying at the open reverses during the day. Behaviour matters for intraday continuation only if it keeps generating flow.
3. **Liquidity provision makes reversal the default** for individual securities at lags under an hour. It is the baseline every intraday momentum claim must beat.
4. **Metaorders create continuation, but others see it first.** The window in which order imbalances predicted NYSE returns shrank from thirty minutes to five between 1996 and 2002.
5. **Price-contingent flows drive intraday momentum.** Leveraged ETFs trade $A\,\ell(\ell-1)R$ in the direction of the move. Short-gamma dealers, stops and hedgers of equity portfolios do the same, on a schedule. Intraday momentum is a property of the current flow composition and is not constant.
6. **The main document's momentum earns its return overnight.** In US stocks, momentum profits accrue between the close and the open. During the session, momentum stocks have underperformed.

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3. The signature: autocorrelation inside the day

The main document's second leg is an observable diagnostic, the variance-ratio profile. It shows whether a market trends at a given horizon before the trader builds any signal. This section rebuilds that diagnostic for a process that is not stationary within the day.

3.1 Why the stationary diagnostics break

Three features of intraday data prevent a single autocorrelation function from describing it.

The intraday profile. Volatility, volume and spreads follow a U shape across the session, or a reverse J in which the open dominates. Wood, McInish & Ord (1985) and Harris (1986) documented the pattern in NYSE transaction data. Admati & Pfleiderer (1988) explain the concentration. Traders who can choose when to trade cluster together to reduce their costs, and informed traders follow them. Andersen & Bollerslev (1997) show that the profile badly distorts naive statistics. If left in, it dominates the autocorrelations of absolute returns and obscures the slow decay of volatility persistence. If filtered out, the underlying dynamics appear. The distortion of signed returns is smaller but real. A lag-one autocorrelation pooled over all five-minute bars mostly reflects the first and last half hour, because those bars carry most of the variance.

The overnight bar. The previous close to the open is one "bar" spanning seventeen and a half hours of calendar time and a large, variable share of the day's variance. It is a different random variable from any intraday bar, and a rolling window that crosses it mixes the two.

The timetable. Flows arrive at fixed times: the opening auction, scheduled announcements at 8:30 and 10:00 a.m. and 2:00 p.m., the closing auction, month-end fixes and option expiries. Their effects occur at particular times of day, not at particular lags.

A process whose mean and autocovariance repeat with a fixed period is *periodically correlated*, or *cyclostationary*. Intraday returns are, to a first approximation, periodically correlated with a period of one day. Diagnostics for such a process must be indexed by time of day.

3.2 The periodic autocovariance surface and the intraday P&L identity

Section 5.2 of Trend-Following proves that the expected P&L of a linear trend rule is a weighted sum of return autocovariances. The intraday version needs one more index.

Let a rule decide its position at the end of bar i from the returns so far that day, including the overnight return as bar 0:

$$x_{d,i} \;=\; \sum_{j=0}^{i} w_{i,j}\, r_{d,j}, \qquad r_{d,0} \equiv r_d^{\text{on}}.$$

The weights $w_{i,j}$ depend on *both* the decision time i and the bar j. For a session-anchored rule they reset every day. The rule holds $x_{d,i}$ over bar $i+1$, so its daily P&L is $\sum_i x_{d,i}\, r_{d,i+1}$. Its expectation is

$$\mathbb{E}\Big[\sum_i x_{d,i}\, r_{d,i+1}\Big] \;=\; \sum_i \sum_{j\le i} w_{i,j}\; \mathbb{E}\big[r_{d,j}\, r_{d,i+1}\big] \;=\; \sum_i \sum_{j\le i} w_{i,j}\,\Big[\gamma(i{+}1,\; i{+}1{-}j) \;+\; \mu_j\, \mu_{i+1}\Big],$$

where $\gamma(i,k)$ is the periodic autocovariance and μ_i is the mean return of bar i across days. The identity states:

> **The expected P&L of any linear intraday rule is the inner product of its weight matrix with the periodic autocovariance surface, plus a term from time-of-day mean returns.**

Four readings of the identity show how to use it.

- **Gao and co-authors' timing rule** puts weight only on the overnight bar and the first half hour, and holds a position only in the last half hour. It samples a single cell of the surface, the last half hour against the overnight-plus-first-half-hour return. Its sign-rule version is a nonlinear function of the same cell.
- **Baltussen and co-authors' rule** puts equal weight on every bar from the previous close to 3:30 p.m. It samples the whole column of the surface that ends at the last half hour.
- **A rolling five-minute momentum rule** puts its weight on the cells next to the diagonal, at every time of day. An example is the sign of the last hour's return, re-evaluated every bar. Bid–ask bounce and inventory make those cells negative (§3.3). The rule bets on the wrong part of the surface.

- **The drift term** $\mu_j \mu_{i+1}$ is the intraday version of the drift confound that section 5.3 of Trend-Following warns about. A rule that is usually long late in the day earns the average late-day return whether or not anything is predictable. In Gao and co-authors' SPY sample, being long in the last half hour earned −1.11% a year. So the drift term worked *against* the timing rule there.

Two practical consequences follow. First, **estimate the surface before designing rules**. One regression per cell is cheap, and it shows where any rule's weights ought to sit. Second, **many rules that look different are different weightings of the same few cells**. This is the intraday version of the main document's equivalence results (§5.2).

3.3 What the surface looks like

No single paper estimates the whole surface, but the literature estimates enough cells to sketch it. The slopes quoted are predictive-regression coefficients. A slope of 0.042 means that a 1% move in the predictor window forecasts a 4.2 basis-point move in the target window.

Target ← predictor	Sign and size	Where	Source
Bar ← previous bar, 1–5 minutes, transaction prices	Negative; about −0.05 at one minute for a 5 bp spread	Single stocks	Roll (1984); §3.6
Bar ← previous bar, 5 minutes, liquid stocks	About zero	NYSE stocks	Chordia, Roll & Subrahmanyam (2005)
Half hour ← earlier half hours, same day	Negative at short lags	US stocks	Heston, Korajczyk & Sadka (2010)
Last half hour ← rest of day	Positive; slope 0.042, R^2 2.45%	Equity index futures	Baltussen, Da, Lammers & Martens (2021)
	Slope 0.019, R^2 0.64%	Bond futures	same
	Slope 0.012 and 0.007, R^2 0.15–0.19%	Commodity and currency futures	same
Last half hour ← overnight plus first half hour	Positive; slope 0.069, R^2 1.6%	SPY	Gao, Han, Li & Zhou (2018)
Last half hour ← second-to-last half hour	Positive in equities and bonds, not in commodities or currencies	Futures	Gao et al. (2018); Baltussen et al. (2021)
Last half hour today ← last half hour yesterday	Negative in equity index futures, positive in currency futures	Futures	Baltussen et al. (2021)
Next 1–3 days ← last half hour	Negative: the move partly reverses	Futures	Baltussen et al. (2021)
Same half hour, 1 to 40+ days later	Positive	US stocks, cross-section	Heston, Korajczyk & Sadka (2010)
Afternoon session ← morning session	Positive	Chinese stocks, across the lunch break	Zhang, Ma & Zhu (2019)

The surface is negative along a band next to the diagonal and close to zero in most of the interior. It is positive in one column (the last half hour) and at one set of long lags (the same time on later days, in the cross-section only). Intraday momentum depends on *where the target window sits in the day*. It does not depend on the lag.

3.4 The variance ratio, one day at a time

The main document computes the variance ratio as an average over many windows. Inside the day, a version computed for *each day* is more informative:

$$\mathrm{VR}_d \;=\; \frac{\left(\sum_{i=1}^{M} r_{d,i}\right)^2}{\sum_{i=1}^{M} r_{d,i}^2} \;=\; 1 \;+\; \frac{2\sum_{1\le i<j\le M} r_{d,i}\, r_{d,j}}{\sum_{i=1}^{M} r_{d,i}^2}.$$

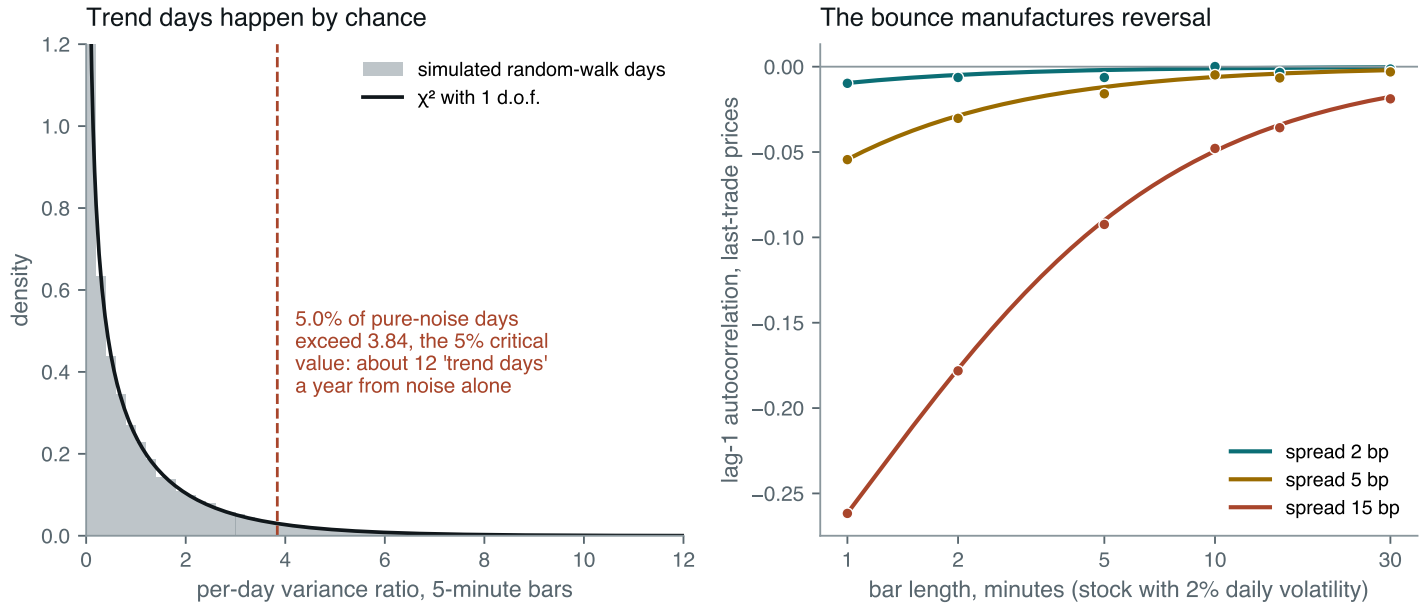
The sum runs over the day's M intraday bars only. It excludes the overnight bar 0, which §3.1 treats separately as a different random variable. The numerator is the square of the day's net move. The denominator is the day's realised variance. The ratio exceeds 1 exactly when the day's bars are, on balance, positively correlated with one another, which means the day trended. It falls below 1 when the bars are, on balance, negatively correlated, which means the day reversed repeatedly. Its average across days estimates one plus twice the variance-weighted average within-day autocorrelation. This is the one-day variance ratio, measured with intraday bars.

Its null distribution is known and corrects intuition. Suppose the day is a driftless random walk with *any* deterministic intraday volatility profile. Write $v_d = \sum_{i\ge 1} \mathrm{Var}(r_{d,i})$ for the variance of the session's net move. Then that net move is normal with variance v_d. The realised variance $\sum_{i\ge 1} r_{d,i}^2$ converges to v_d as the bars get finer, which is the property that makes realised variance a consistent estimator. So

$$\mathrm{VR}_d \;\longrightarrow\; \frac{v_d\, Z^2}{v_d} \;=\; Z^2 \;\sim\; \chi^2_1,$$

This holds whatever the day's volatility and whatever the profile. The 5% critical value of χ^2_1 is 3.84, which corresponds to a day whose net move is at least about twice its realised volatility. On pure noise, **one day in twenty** exceeds that value, or about twelve days a year.

The simulation behind the figure checks this with persistent day-to-day volatility, fat-tailed shocks and a U-shaped profile. With five-minute bars, 5.0% of noise days exceed 3.84. With one-minute bars, 5.1% do. Practitioners' *trend days* are clean one-way sessions that close near their extreme. At that frequency, noise produces such days. A trader who takes them as evidence of intraday momentum is observing the tail of a chi-squared distribution.



Three uses follow.

- **The right diagnostic is the whole distribution.** Compare the empirical distribution of VR_d with χ^2_1. A mean above 1 and excess mass in the upper tail indicate within-day persistence. A mean below 1 indicates within-day reversal. Conditional versions carry the most information. Examples are versions on high-volatility days, on days when dealers are short gamma and on announcement days.
- **Prefer it to the efficiency ratio.** Section 4.5.3 of the main document describes Kaufman's efficiency ratio, $|\sum r| / \sum |r|$. Inside the day it depends on the bar size even on pure noise. The same simulation gives a mean of 0.118 with five-minute bars and 0.056 with one-minute bars. A threshold tuned at one bar size has no meaning at another. The variance ratio is scale-free under the null.
- **Measure it on midquotes.** With transaction prices at fine sampling, bid–ask bounce inflates the realised variance in the denominator but not the net move in the numerator. This pushes VR_d below 1 and creates false "reversal days". Use midquotes or five-minute sampling, as with realised variance generally (Appendix A.4 of the main document).

3.5 Choosing the clock

A clock defines what counts as one unit of time. The unit can be a minute, a trade, a fixed amount of volume or a fixed amount of variance. The choice matters because intraday returns look very different under different clocks.

Clark (1973) modelled returns as a process running on a random "business time" driven by trading activity. This explains fat tails in calendar time as a mixture of normals. Ané & Geman (2000) find that returns are close to normal in transaction time, which counts trades rather than minutes. Sampling in volume or trade time largely removes the U shape, because volume and volatility move together. Easley, López de Prado & O'Hara (2012) popularised the *volume clock* for measuring order-flow toxicity. Their toxicity measure itself is **[Contested]**. Andersen & Bondarenko (2014) find that it predicts short-run volatility poorly. It peaked after the 2010 Flash Crash, not before. Its apparent predictive content is largely a mechanical consequence of trading intensity.

Two methods remove the profile, and they suit different problems.

1. **Re-clock.** Build bars of equal volume or an equal number of trades. This suits microstructure and flow work, where activity is the natural unit.
2. **Rescale.** Keep clock-time bars and divide each return by an estimate of its standard deviation, $\hat{\sigma}_d \, \hat{s}_i$. This is the multiplicative decomposition of Andersen & Bollerslev (1997), which writes volatility as a day-level volatility times a

periodic intraday factor.

For intraday momentum, I recommend the second method. The flows behind the effect follow a schedule in clock time. The close is at 4:00 p.m. however much has traded. So the bars should stay in clock time, and the *normaliser* should carry the profile. §5.3 gives the practical version.

3.6 Measurement artefacts that look like momentum or reversal

Intraday data creates false patterns. Six are common enough that every pipeline needs a test for them (§7.7).

1. **Bid–ask bounce.** If each bar closes on a buy or a sell at random, transaction-price returns acquire a lag-one autocovariance of $-s^2/4$ for a spread s. This is the result of Roll (1984), which section A.2 of the main document derives. The *autocorrelation* is $-(s^2/4)/(\sigma_b^2 + s^2/2)$, where σ_b is the bar's true volatility, so it grows as bars shrink. For a stock with 2% daily volatility and a 5 basis-point spread it is −0.054 at one minute and −0.012 at five. With a 15 basis-point spread, it is −0.26 at one minute, −0.09 at five and −0.02 at thirty (right panel of the figure above). In the same simulation, a one-minute reversal rule shows a profit on transaction prices. The profit is 0.48 basis points per bar with a 5 basis-point spread and 3.2 with a 15 basis-point spread. On midquotes the profit is exactly zero. The backtest earns part of a spread that a real trader would pay.
2. **Stale prices in a cash index.** An index computed from last trades mixes fresh and stale prices whenever its constituents trade at different times. This induces spurious positive autocorrelation (Lo & MacKinlay, 1990). The open is the extreme case. At 9:30 nothing has traded, so the index's first print is essentially yesterday's close, and the overnight gap enters the index gradually as constituents open. The simulation behind the figure makes a stylised assumption that constituents open on average two minutes late. Then only 39% of the overnight gap is visible after one minute and 92% after five. The correlation between the index's 9:30–9:35 change and its 9:35–10:00 change comes out at 0.18, against −0.02 for the true index. This is a spurious opening "momentum" with an R^2 of about 3%, double the 1.6% of the best-known intraday momentum result, produced by staleness alone. Use futures or ETFs, or rebuild the index from its constituents' actual opening prices. The same staleness makes the cash index's official open-to-close return contain most of the overnight return.
3. **Auction prints.** Official opening and closing prices come from auctions, whose prices can deviate from the continuous market and then revert (Bogousslavsky & Muravyev, 2023). Mixing auction prints with continuous-trading prices creates a bar with its own dynamics. Decide whether the last half hour ends at the auction price, which is tradable with market-on-close orders, or at the last continuous trade, and be consistent.
4. **Timestamps.** Several timestamp problems misalign bars by one or more periods. Some data labels bars by their start time rather than their end time. Exchange and consolidated-feed timestamps may differ. Daylight-saving changes fall on different dates in the US and Europe. Futures sessions start the previous evening. An off-by-one bar is look-ahead (§7.3).
5. **The Epps effect.** Correlations measured with high-frequency returns shrink towards zero as the sampling interval falls, because assets trade asynchronously (Epps, 1979). At the extreme, Budish, Cramton & Shim (2015) show that the correlation between ES and SPY breaks down at horizons of milliseconds. Intraday betas are therefore biased towards zero. Residual returns then contain market moves that the beta missed (§5.6).
6. **Universe selection with later information.** A universe chosen using the whole day's volume or range, such as "stocks that gapped and ran", uses information that did not exist at the time of the trade (§7.3).

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§3 Key takeaways

1. **Intraday returns are periodically correlated, not stationary.** The U-shaped profile, the overnight bar and the schedule of flows make any single autocorrelation or variance ratio a mixture of different random variables.
2. **The expected P&L of a linear intraday rule is its weight matrix dotted into the periodic autocovariance surface**, plus a time-of-day drift term. Estimate the surface first. Most rules are different weightings of the same few cells.
3. **The surface is negative next to the diagonal and positive in the last-half-hour column.** Intraday momentum is a property of where the target window sits in the day, not of the lag.
4. **Measure the variance ratio per day.** On noise it is chi-squared with one degree of freedom regardless of the volatility profile, so one day in twenty is a "trend day" by chance. Test the whole distribution, conditionally.
5. **Keep the clock, carry the profile in the normaliser.** Scheduled flows occur in clock time. Volatility follows business time.

6. **Audit for artefacts before believing any pattern.** Bid–ask bounce creates reversal of −0.26 at one minute for a 15 basis-point spread. A stale cash index creates opening momentum with an R^2 of about 3%. Timestamp slips create look-ahead.

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4. The evidence, effect by effect

The same fields, in the same order, describe each effect below, so a reader can compare the effects directly. Shorter entries combine several fields into one. The fields are:

- **Claim** — what predicts what.
- **Construction** — predictor, target, universe and sample.
- **Magnitude** — the headline numbers, before costs unless stated.
- **Mechanism offered** — what the authors think causes it.
- **Replications and critiques** — who has checked it, and what they found.
- **Tradability** — whether it survives costs, and for whom.
- **Verdict** — my assessment, tagged.

4.1 Market intraday momentum: the first half hour predicts the last

Claim. The market's return from the previous close through the first half hour of trading predicts its return in the last half hour.

Construction. Gao, Han, Li & Zhou (2018) use SPY from 1993 to 2013, split into thirteen half hours. The first half-hour return is measured from the previous close, so it includes the overnight move. They also use the second-to-last half hour as a predictor, and repeat the analysis on ten other heavily traded ETFs and two international index futures.

Magnitude. In the working-paper version the slope is 0.069 and the R^2 is 1.6%. The second-to-last half hour alone gives an R^2 of 1.1%, and the two predictors together give 2.6%. The out-of-sample R^2 is 1.2% for the first predictor and 1.8% for both. This statistic compares forecasts made only with data available at the time against the historical average. It turns negative when the forecasts do worse (A.19). A timing strategy goes long or short in the last half hour according to the sign of the first half-hour return. It earns 6.67% a year before costs, with a volatility of 6.19% and a Sharpe ratio of 1.08. Simply being long in the last half hour earns −1.11% a year. The strategy's daily returns have a skewness of 0.90 and a kurtosis of 15.65.

The effect is heavily state-dependent. Sorted by first half-hour volatility, the R^2 is 0.6% (and insignificant) in the calmest third of days, 1.0% in the middle third and 3.3% in the most volatile third. It is stronger on high-volume days, in recessions and on days with major macroeconomic releases. In the financial crisis from December 2007 to June 2009, the first predictor's R^2 reaches 4.1%, and the two predictors together reach 6.9%. Excluding those days, they fall to 0.8% and 1.1%.

Mechanism offered. The authors offer two mechanisms. The first is infrequent rebalancing by some institutions at the start and end of the day (Bogousslavsky, 2016). The second is late-informed traders who act near the close on information that arrived in the morning.

Replications and critiques.

- *International.* Li, Sakkas & Urquhart (2022) find significant in-sample predictability in 12 of 16 developed equity markets in 2005–2017. The predictability is stronger during the financial crisis, when the US slope is about four times larger than outside it. It is also stronger when liquidity is low, volatility is high and information is discrete. Out of sample, the evidence is thinner than the paper's summary suggests. Only 5 of the 16 markets have a positive out-of-sample R^2. Adjusted tests of forecast improvement reject the null in 10 markets, and a forecast-encompassing test rejects it in 14.
- *Other markets.* The pattern appears in Chinese stocks, including the morning session predicting the afternoon across the lunch break (Zhang, Ma & Zhu, 2019). It appears in four Chinese commodity futures, strongest after high-volume, high-volatility opening sessions (Jin, Kearney, Li & Yang, 2020). It appears in the United States Oil Fund in 2006–2018. There only the first half hour predicts, its overnight component carries most of the information, and scheduled inventory announcements add nothing (Wen, Gong, Ma & Xu, 2021). It appears in the rouble–dollar exchange rate in 2005–2014. There the authors reject informed trading in favour of liquidity providers' aversion to holding positions overnight (Elaut, Frömmel & Lampaert, 2018). It appears in bitcoin, which has no close, using trading volume to define a session (Shen, Urquhart & Wang, 2022). Wen, Bouri, Xu & Zhao (2022) find both momentum and reversal across cryptocurrencies.
- *Critiques.* Rosa (2022) finds that the predictability *disappears* out of sample. A Markov-switching model identifies two regimes in which predictability depends on the strength of the signal. A strategy that trades only beyond a threshold

beats one that is always active. Pooling 17 equity index futures over 1974–2020, Baltussen, Da, Lammers & Martens (2021) find that the first-half-hour predictor has an in-sample R^2 of 1.49% and an out-of-sample R^2 of **−1.71%**. It forecasts worse than the historical mean. When the first half hour and the rest of the day disagree in sign, the first half hour's coefficient has the *wrong* sign.

Tradability. In SPY or ES, costs are small relative to an average edge of about 2.6 basis points a day (§6.2). Capacity is small (§6.5).

Verdict. **[Fact]** that the pattern appears in-sample in many markets, and that it is strongest in volatile periods. **[Contested]** as a stable, unconditional effect. The published out-of-sample record is weak, and the headline results depend on a small number of crisis days. Excluding eighteen months of 2007–2009 halves the R^2. My view is that the first half hour is a noisy proxy for the stronger rest-of-day predictor in §4.2. Do not trade it unconditionally.

4.2 The rest of the day predicts the last half hour: the hedging channel

Claim. The return from the previous close to 30 minutes before the close predicts the last half-hour return, across asset classes.

Construction. Baltussen, Da, Lammers & Martens (2021) use more than 60 futures from 1974 to 2020: 17 equity index, 16 government bond, 21 commodity and 8 currency contracts. They use each market's "common" trading hours and pooled regressions with standard errors clustered by time and by market.

Magnitude.

Asset class	Slope	R^2	Out-of-sample R^2	Timing Sharpe, before costs	Success rate
Equity index futures	0.042 (t = 7.3)	2.45%	2.88%	1.73	55%
Government bond futures	0.019	0.64%	0.60%	1.62	55%
Commodity futures	0.012	0.15%	0.07%	1.42	56%
Currency futures	0.007	0.19%	0.26%	0.87	53%

The Sharpe ratios are for equal-weighted portfolios of the contracts in each class, which diversify across markets. Split in two periods, the equity slope falls from 0.060 (1974–1999) to 0.040 (2000–2020). The bond slope falls from 0.032 to 0.015, and the commodity slope from 0.016 to 0.010. In currencies the effect is insignificant after 2000. The authors report a positive Sharpe ratio net of costs for S&P 500 futures at a cost of one tick.

Mechanism offered. Hedging demand from participants who are short gamma, above all options market makers and leveraged ETFs. Four pieces of evidence support it. The effect is present in the S&P 500 when a proxy for dealers' gamma is negative and grows as it becomes more negative. Leveraged-ETF hedging demand predicts the size of the effect across indices and over time. The last half-hour move reverses over the following one to three days, significantly in equities, bonds and commodities. This is what price pressure predicts and information does not. Finally, the predictability does not extend beyond 4:00 p.m. into the fifteen minutes during which S&P 500 futures keep trading after the cash close.

Replications and critiques.

- Ivanov & Lenkey (2018) argue that leveraged-ETF rebalancing, net of investors' capital flows, has an economically insignificant effect on late-day returns.
- Barbon & Buraschi (2020) link the *sign* of intraday autocorrelation to dealers' gamma. Momentum occurs when dealers are short gamma, and reversal when they are long. Section 7 of Dealer Hedging covers the mechanics and the zero-day-option debate.
- Gao and co-authors' infrequent-rebalancing and late-informed channels predict the same sign, so the channels are not mutually exclusive.

Tradability. For equity index futures the edge comfortably exceeds costs (§6.2). For bond, commodity and currency futures, whose slopes are a third to a sixth as large, the edge is close to or below the cost of crossing the spread. For these markets the effect is better used to time trades than as a stand-alone strategy.

Verdict. **[Fact]** that the rest-of-day predictor works in-sample across four asset classes and out of sample in equities and bonds, and that it beats the first-half-hour predictor. **[Contested]** as to how much of it hedging explains. **[Fact]** that it has weakened since 2000 outside equities. This is the best-documented intraday momentum effect. I would build on it.

4.3 Same-time-of-day continuation in the cross-section

Claim. A stock's return in a given half hour predicts its return in the same half hour on subsequent days.

Construction. Heston, Korajczyk & Sadka (2010) use US stocks and half-hour returns.

Magnitude. Returns continue at lags that are exact multiples of a trading day, for at least 40 trading days. Short-term reversal occurs alongside this continuation within the day. Volume, order imbalance, volatility and spreads show similar periodic patterns but do not explain the return pattern. The authors estimate that timing trades to this periodicity can reduce execution costs by the equivalent of the effective spread (A.1).

Mechanism offered. Institutions trade at the same time of day on consecutive days, as they work a large order or as a fund's flows recur. Bogousslavsky (2016) formalises the link, and Murphy & Thirumalai (2017) connect the pattern to repetitive institutional net order activity.

Replications and critiques. Baltussen, Da, Lammers & Martens (2021) find that the effect does not aggregate to the index. In equity index futures, yesterday's last half-hour return predicts today's *negatively*. A cross-sectional seasonality removes the market. A time-series effect is mostly the market. They are different objects.

Tradability. As a stand-alone long–short strategy the per-trade return is small relative to single-stock costs. As an execution tool it is valuable.

Verdict. [Fact] for the pattern in US data. **[Hypothesis]** for the mechanism. Treat it as information about *when to trade*, not *what to trade*.

4.4 Where the main document's momentum is earned: overnight

Claim. In US stocks, the profits of twelve-month momentum accrue overnight, and the intraday component has the opposite sign.

Construction. Lou, Polk & Skouras (2019) decompose close-to-close returns of US stocks and fourteen trading strategies into close-to-open and open-to-close components.

Magnitude. Momentum and reversal strategies earn their profits entirely overnight. Most other strategies earn theirs entirely intraday. Typically, their profits in the other component have the opposite sign. Firm-level returns show continuation within each component and reversal across them. The pattern lasts for years.

Mechanism offered. Opposing flows from two clienteles. Individuals tend to trade near the open. Institutions trade during the day. The smoothed spread between a strategy's overnight and intraday components forecasts its future close-to-close performance.

Replications and critiques. Bogousslavsky (2021) finds that size and illiquidity premia accrue in the last thirty minutes. Profitability and idiosyncratic-volatility anomalies accrue during the day and lose overnight. This is consistent with mispricing at the open. Hendershott, Livdan & Rösch (2020) find that market risk is priced overnight and priced negatively during the day. Akbas, Boehmer, Jiang & Koch (2022) show that the intensity of the daily opposing flows predicts future returns. The standing objection is measurement. Opening prices are noisier than closing prices.

Tradability. This is not a strategy in itself. It constrains anyone trading momentum stocks within the day. It also matters for when a slower momentum book trades (§8.2).

Verdict. [Fact] for the decomposition in US data. **[Contested]** for the mechanism.

4.5 Intraday cross-sectional momentum: mostly absent

Claim. Ranking stocks on their past-hour returns and buying the winners is profitable. This is Jegadeesh and Titman's portfolio applied at an intraday horizon.

Evidence against. For German blue chips, Herberger, Horn & Oehler (2020) find no momentum at 15–60 minute horizons. They find strong but economically insignificant reversal at 60–300 minutes. Heston, Korajczyk & Sadka (2010) find reversal at short lags within the day in the US. Chordia, Roll & Subrahmanyam (2005) find liquid NYSE stocks weak-form efficient at five minutes. Past returns alone predict nothing.

Verdict. [Fact] Reversal dominates the intraday cross-section at lags under an hour. Outside the same-time-of-day seasonality of §4.3, no robust evidence of cross-sectional intraday momentum exists. The main document's cross-sectional momentum translates least well here.

4.6 Levels, stops and fixes

Claim. Prices continue after crossing certain levels, and drift predictably into scheduled benchmark fixes.

Evidence.

- *Round numbers and stops.* Using a large dealer's foreign-exchange order book, Osler (2003) shows that take-profit orders cluster at round numbers, where trends tend to reverse. The paper also shows that stop-loss orders cluster just beyond round numbers, where trends accelerate once crossed. Osler (2005) documents the resulting price cascades. Exchange rates move unusually fast on reaching levels where stop-losses cluster. The response to stops is larger and longer-lasting than the response to take-profits.
- *Fixes.* Before the month-end London 4 p.m. fix, a market's relative equity appreciation predicts its currency's depreciation, with partial reversal the next day (Melvin & Prins, 2015). Across 21 currency pairs in 2000–2013, rate changes into the fix and out of it are negatively correlated (Evans, 2018). At the Tokyo fix, the data show two biases (Ito & Yamada, 2017). Customer orders are predictably biased towards buying foreign currency. Before 2008, the fixing rate set by banks was biased upwards.

Tradability. Stop cascades are fast and gappy. The rate can jump over levels without trading, so the continuation after a break is hard to capture. Fix flows are predictable in sign. The pre-fix drift is partly offset by the post-fix reversal. Regulators have fined banks for trading around the fix.

Verdict. **[Fact]** for the order clustering and the fix patterns in the samples studied. **[Hypothesis]** that stop cascades explain breakout continuation in equities. No comparable order-book study of equities exists in the public literature.

4.7 Technical rules on intraday bars

Claim. Standard technical rules are profitable on intraday data. These rules include moving averages, channels, oscillators and rules found by search.

Evidence.

- Neely & Weller (2003) search for intraday rules on half-hourly exchange rates in 1996. They use a genetic program (an automated search that breeds and mutates candidate trading rules) and an optimised linear forecasting model. Once realistic transaction costs and trading hours apply, neither method earns excess returns. The rules do find "remarkably stable patterns" in the data.
- Marshall, Cahan & Cahan (2008) test 7,846 popular technical rules on five-minute data for an S&P 500 ETF in 2002–2003. They use bootstrap tests and a correction for data snooping. None of the rules is profitable.
- Schulmeister (2009) tests 2,580 technical models on the S&P 500. On daily data, profitability has declined steadily since 1960 and turned negative in the early 1990s. On thirty-minute data, the same models earn an average *gross* return of 7.2% a year from 1983 to 2007, with no clear downward trend.
- A survey by Neely & Weller (2012) reports a tick-data rule for sterling–dollar. The rule was profitable after costs in 2003 and not in 2008.

Verdict. **[Fact]** Generic technical rules on intraday bars do not survive realistic costs and corrections for data snooping. Where gross profits appear, they are in index futures, before costs. The cost wall says any residual edge would be in the same place.

4.8 Practitioner systems: opening-range breakouts, stocks in play and the noise area

These are the most widely circulated intraday momentum strategies, and the least independently verified.

Claim. Taking a position when the price leaves a range set early in the session, and holding it to the close, earns large risk-adjusted returns.

Construction and magnitude.

- *The opening-range breakout (ORB)* is a practitioner rule usually credited to Crabel (1990). Go long if the price breaks above the high of the first N minutes. Go short if it breaks below the low. Exit at the close. Holmberg, Lönnbark & Lundström (2013) test a version on crude oil futures in 1983–2011. They use only daily open, high, low and close prices, with entry thresholds set from the sample's return distribution. Average returns are about 0.20–0.26% per trade, and success rates are near 61%. The test assumes zero commissions and chooses thresholds on the full sample.
- Zarattini & Aziz (2023) run a five-minute ORB on QQQ in 2016–2023. They report an annualised alpha of 33% net of commissions. They also report 1,484% cumulative return when the strategy holds the positions in a three-times leveraged ETF.

- Zarattini, Barbon & Aziz (2024a) extend the five-minute ORB to more than 7,000 US stocks in 2016–2023. The strategy enters on a stop order at the opening-range extreme, in the direction of the first five-minute bar. The stop loss is 10% of the 14-day average true range, a measure of the typical daily trading range (section 4.4.4 of the main document). Across all eligible stocks the result is poor. The return is 3.2% a year at 6.6% volatility, a Sharpe ratio of 0.48 against 0.78 for the S&P 500. The paper also restricts the strategy to the twenty *stocks in play* with the highest relative volume in the first five minutes. Stocks in play are stocks with unusual activity, mostly on news. For these stocks it reports a net cumulative return above 1,600% and a Sharpe ratio of 2.81.
- Zarattini, Aziz & Barbon (2024b) propose the *noise area* for SPY in 2007–early 2024. For each time of day, the band's half-width is the average absolute move from the open at that time over the previous 14 days. Any overnight gap widens the band on one side. Every half hour, the strategy goes long above the band and short below it. It exits at the close or on crossing to the other side of the band. The paper assumes commissions of $0.0035 per share and slippage of $0.001. It reports that the base version earns 6.2% a year at a volatility of 10.9%, with a Sharpe ratio of 0.61. Adding a trailing stop at the volume-weighted average price (VWAP) raises the Sharpe ratio to 1.24 (9.7% a year at 7.7% volatility). Adding volatility-targeted sizing gives 19.6% a year at 14.3% volatility, a Sharpe ratio of 1.33, a hit rate of 43% and a maximum drawdown of 25%. The sizing aims at 2% daily volatility and caps leverage at four. All of these are the paper's own figures. Its Sharpe ratios are not exactly the reported return divided by the reported volatility, so treat the second decimal with care.

Mechanism offered. Persistent imbalances between buyers and sellers become visible once the price leaves the range that "normal" noise would produce. For stocks in play, the mechanism is news and attention.

Replications and critiques.

- *Interested parties.* The three Zarattini papers are working papers in the Swiss Finance Institute series. They are not peer-reviewed. Andrew Aziz is the author of the day-trading guide *How to Day Trade for a Living*. He runs the Bear Bull Traders day-trading community and founded a proprietary trading firm. Carlo Zarattini runs Concretum Group, a quantitative research firm. Andrea Barbon is an academic at the University of St. Gallen. Independent replication is the standard to wait for. These affiliations do not make the results wrong.
- *The noise area on noise.* I ran the base noise-area rule, without the gap adjustment, on 20,000 simulated random-walk days with a realistic intraday profile. The figure in §3.4 uses the same simulation. The price is outside the band at **42%** of the half-hourly checks, and the rule trades on 86% of days. The band's construction implies this result. Under normality, the absolute move from the open exceeds its own mean with probability $P(|Z| > \sqrt{2/\pi}) \approx 0.42$. The band is a threshold at about 0.8 standard deviations. It does not filter for "abnormal" imbalance. On noise, the rule's gross P&L is zero, as it must be. Everything in the reported results comes from the data, and from the choices the authors made while studying it.
- *Degrees of freedom.* The reported Sharpe ratio rises from 0.61 to 1.24 when the authors add the VWAP stop, and to 1.33 with sizing. Each step uses the same seventeen years. The 14-day window, the half-hourly decision times, the gap adjustment, the VWAP stop, the 2% target and the leverage cap are all choices. A follow-up working paper already optimises the parameters and exits further (Maróy, 2025). The headline 19.6% a year is mostly leverage of two to four times applied to the 1.24-Sharpe version.
- *In its favour.* Seventeen years is a long sample. At a Sharpe ratio of 1.33, the result is far too large for selection alone to produce from any realistic number of trials (§7.2). The rule's mechanism also matches the academic evidence. It is a thresholded, early-entry version of the rest-of-day effect of §4.2. Thresholds help precisely because the effect is concentrated on large-move, high-volatility days (Gao and co-authors; Rosa).
- *Stocks in play.* The universe definition is the central issue, and the paper defines it correctly. Relative volume in the first five minutes is known at 9:35, before any entry. The cost model is the weakest part. Costs are a commission of $0.0035 a share on a $25,000 account. I find no allowance for the spread or for slippage on stop-order entries in fast-moving stocks. That cost matters most for this strategy (§6.2). The step from a Sharpe ratio of 0.48 for all stocks to 2.81 for the top twenty is also a selection made on the same eight years.

Tradability. Uncertain. The base SPY rule's Sharpe ratio of 0.61 net of the assumed costs is a more defensible figure to rely on than the headline 1.33.

Verdict. [Contested]. The mechanism is plausible, and consistent with the rest-of-day effect. Discount the magnitudes heavily until someone without a stake in the result replicates them out of sample.

4.9 The scorecard

Effect	Best evidence	Out of sample	Trend over time	Status
Rest of day → last half hour, index futures	60+ futures, 1974–2020	Positive in equities and bonds	Weaker since 2000, most outside equities	**[Fact]** exists; **[Contested]** cause
First half hour → last half hour	SPY 1993–2013; 16 markets	Mixed to negative	Concentrated in crises	**[Contested]**
Same half hour on later days	US stocks	Not a strategy; an execution tool	—	**[Fact]** pattern
Momentum earned overnight	US stocks, decades	Persistent for years	—	**[Fact]** pattern, **[Contested]** cause
Intraday cross-sectional momentum	DAX, NYSE	Absent; reversal instead	—	**[Fact]** absent
Stop cascades, fixes	FX order book; 21 currency pairs	—	Fix practices reformed	**[Fact]** in FX
Generic technical rules on intraday bars	7,846 rules; 2,580 models	Fail after costs and snooping corrections	Gross profits persisted in index futures to 2007	**[Fact]** fail
Opening-range and noise-area systems	Working papers by interested parties	Not independently replicated	—	**[Contested]**

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§4 Key takeaways

1. **The best-documented effect is the rest of the day predicting the last half hour**, in 60+ futures over 1974–2020. In equity index futures the slope is 0.042 and the out-of-sample R^2 is 2.9%. Partial reversal follows over the next days.
2. **The first-half-hour version is weaker than its reputation.** Out of sample it fails in pooled equity futures (R^2 −1.7%) and in 11 of 16 international markets. In SPY, its R^2 halves when eighteen months of 2007–2009 are removed.
3. **Intraday momentum is strongest when it is hardest to trade.** It is strongest on volatile, high-volume, news and crisis days. Its returns have a heavy right tail (kurtosis 15.7 in SPY timing returns).
4. **The effect has weakened since 2000** in bonds, commodities and currencies. Evidence that capital flows offset leveraged-ETF rebalancing contests its hedging explanation.
5. **The evidence shows no robust intraday cross-sectional momentum.** Reversal dominates below an hour. The main document's momentum is earned overnight.
6. **Generic technical rules on intraday bars fail** realistic costs and snooping corrections. None of the 7,846 rules tested on five-minute S&P 500 data was profitable.
7. **Practitioner breakout systems report large numbers, but no independent study has replicated them.** On pure noise, the price crosses the noise-area rule's band at 42% of checks. Its base Sharpe ratio of 0.61 is the figure to use.

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5. Translating the main document's methods

This section takes each concept of the main document in turn and asks what happens to it inside the day. Each entry has the same five fields:

- **In the main document** — the concept, in one line, with a link.
- **Inside the day** — what changes.
- **Evidence** — what the literature of §4 says.
- **Verdict** — *translates*, *translates with changes*, *inverts*, or *does not apply*.
- **Pitfall** — the characteristic way it goes wrong inside the day.

The table in §5.11 collects the verdicts.

5.1 Lookback, holding period and the skip

In the main document. A lookback L and a holding period H are both counted in bars. The rule skips the most recent month to avoid its opposite-signed reversal (section 1.5 of the main document).

Inside the day. The session replaces the rolling window as the natural unit. The lookback is *anchored*. It starts from the previous close or from the open, not a fixed number of bars back from now. The holding period is a fixed clock window, such as the last half hour, not a fixed number of bars forward. The skip survives in three new forms. Skip the bounce: use midquotes, or leave out the last bar before a decision. Skip the auction print if you cannot trade at it. Treat the overnight return as its own bar with its own weight, not as one more intraday bar.

Evidence. The anchored rest-of-day window beats the shorter first-half-hour window out of sample in pooled equity futures (Baltussen, Da, Lammers & Martens, 2021). Reversal at the shortest lags is universal (§2.3).

Verdict. Translates with changes: rolling becomes anchored.

Pitfall. A rolling lookback that straddles the open mixes the overnight gap into an "intraday" signal. At 9:45, a sixty-minute lookback mostly covers the previous session. A lookback counted in bars also changes meaning with the bar size. Twelve bars is an hour of five-minute bars and twelve minutes of one-minute bars.

5.2 The master form becomes an anchored, time-of-day kernel

In the main document. Almost every momentum signal is a normalised, weighted sum of past returns, $s_t = f(\sum_k w_k r_{t-k}/\hat{\sigma}_t)$. Different indicators are different kernels w_k (section 5.2 of the main document).

Inside the day. The weights depend on the decision time as well as the lag. They reset at the start of each session:

$$s_{d,i} \;=\; f\left(\frac{\sum_{j=0}^{i} w_{i,j}\, r_{d,j}}{\hat{\sigma}_{d,i}}\right),$$

where $\hat{\sigma}_{d,i}$ is the volatility expected for the window being summed at that time of day (§5.3). By §3.2, the rule's expected P&L is the inner product of $w_{i,j}$ with the periodic autocovariance surface.

Evidence. The rules in §4 are all coordinates in this form. This gives a table of equivalences like the one in section 5.5 of the main document.

Rule	As a coordinate in the master form	Exact or approximate
Gao and co-authors' timing rule	Decision at 3:30 p.m.; weight 1 on the overnight bar and the first half hour, 0 elsewhere; f = sign; held for the last half hour	Exact
Baltussen and co-authors' rule	Decision at 3:30 p.m.; weight 1 on every bar since the previous close; f = sign	Exact
Opening-range breakout	An anchored Donchian channel: the signal fires when the price leaves the range of the first N minutes; f is a step with memory	Exact as a channel rule; approximately a thresholded rest-of-day rule
Noise area	The cumulative return from the open, gap-adjusted, divided by the 14-day mean absolute move at that time of day; f is a threshold with hysteresis at the opposite band	Exact
Price minus VWAP	An anchored moving-average displacement in volume time (derived below)	Exact
Moving-average crossover on five-minute bars	A rolling kernel across bars and sessions: weight on the cells next to the diagonal	A different object

VWAP is the intraday indicator most often used as a trend filter, so its table entry gets a short derivation. Let v_j be the volume in bar j. For a decision at bar i, let $\Omega_j = (\sum_{m\le j} v_m)/(\sum_{m\le i} v_m)$ be the share of the volume traded by time i that had already traded by the end of bar j. Work in logs, with $\text{vwap}_{d,i} \approx \sum_{j\le i}(v_j/\sum_{m\le i} v_m)\, p_{d,j}$. Write each price gap as a sum of returns, $p_{d,i} - p_{d,j} = \sum_{k=j+1}^{i} r_{d,k}$. Then swap the order of summation, so that each return $r_{d,k}$ receives the weight of every bar $j < k$. The result is

$$p_{d,i} - \text{vwap}_{d,i} \;\approx\; \sum_{j\le i} \frac{v_j}{\sum_{m\le i} v_m}\,(p_{d,i} - p_{d,j}) \;=\; \sum_{k=2}^{i} \Omega_{k-1}\, r_{d,k}.$$

Each bar's return gets the weight of the share of that volume which traded *before* it. So recent returns get more weight, and the first bar gets none. The result is exactly price minus a simple moving average anchored at the open, computed in volume time. This is the moving-average displacement of section 4.1.5 of the main document, measured in volume time and anchored to the session.

Verdict. Translates with changes. The equivalences transfer. At a matched window and threshold, the ORB, the noise area and the rest-of-day sign rule are close to the same bet.

Pitfall. An anchored sum covers thirty minutes at 10:00 and six hours at 3:30, so its variance grows through the day. Without time-of-day normalisation, a fixed threshold fires mostly late and a sign rule is dominated by late-day noise.

5.3 Volatility normalisation becomes a two-factor normalisation

In the main document. Dividing by volatility is the most valuable transformation the main document describes (section 4.3.1 of the main document).

Inside the day. The normaliser needs two factors: a day-level volatility $\hat{\sigma}_d$ and the intraday profile $\hat{s}_i$. For an anchored window, the expected standard deviation at time i is

$$\hat{\sigma}_{d,0:i} \;=\; \hat{\sigma}_d \Big(\sum_{j=0}^{i} \hat{s}_j^2 \Big)^{1/2}.$$

The sum starts at $j = 0$, the overnight bar, when the window starts at the previous close. It starts at $j = 1$ when the window starts at the open. The day-level factor can use what is known before the window: yesterday's realised variance, a HAR-type forecast, the overnight gap and the range of the first half hour. The profile comes from trailing days. The noise area of §4.8 is a crude estimator of exactly this quantity. Its fourteen-day mean absolute move at each time of day estimates $\sqrt{2/\pi}\,\hat{\sigma}_{d,0:i}$, with no adjustment for today's volatility.

Volatility has a second role inside the day, with no counterpart in the main document. Monthly momentum is normalised by volatility to equalise risk, and volatility is otherwise a nuisance. Inside the day, volatility is also a *state variable* that predicts how strong the effect will be. Gao and co-authors' R^2 is five times higher in the most volatile third of days than in the calmest. So **size positions by volatility, but condition the signal on it**. Dividing the signal by volatility and stopping there discards the most predictive variable in the problem.

Evidence. Gao, Han, Li & Zhou (2018) on volatility terciles. Li, Sakkas & Urquhart (2022) on high volatility and low liquidity. In Zarattini, Aziz & Barbon (2024b), the gain from volatility-targeted sizing is small (a Sharpe ratio of 1.24 to 1.33) next to the gain from the exit rule.

Verdict. Translates, with an added role for volatility: normalise for risk, condition for strength.

Pitfall. Estimating the profile on the full sample is look-ahead. The profile also drifts. The closing auction's share of volume more than doubled between 2010 and 2018, and zero-day options changed the afternoon. So estimate the profile on trailing data.

5.4 Signal transforms and sizing become thresholds and dead bands

In the main document. Sign, rank, clip and squash matter much less than the horizon and the normaliser (section 4.3.4 of the main document).

Inside the day. The transform matters *more*, because the cost of a trade is large relative to its expected edge. A dead band does nothing unless the signal is beyond a threshold. It filters costs as much as signals. It trades only on the days when the expected move is large enough to pay for the round trip. The effect is concentrated on large-move, high-volatility days, so the dead band also selects the days on which the signal is strongest.

Evidence. Rosa (2022) finds that a threshold strategy beats an always-active one. Baltussen, Da, Lammers & Martens (2021) find that trading only when the first half hour and the rest of the day agree in sign raises the Sharpe ratio in equity futures. The Sharpe ratio rises from 1.07 to 1.60, relative to the first half hour alone. The rule skips the days on which the two disagree.

Verdict. Translates, and matters more inside the day than in the main document.

Pitfall. Thresholds are the most-tuned parameters in intraday research. Each one tried is a trial to count (§7.2).

5.5 Time-series versus cross-sectional momentum

In the main document. Time-series momentum (an asset against its own history) and cross-sectional momentum (an asset against its peers) are different products, with different crash risk (sections 4.6.1–4.6.2 of the main document).

Inside the day. The two behave completely differently. Time-series momentum survives at the level of the index. The rest-of-day effect is a market-level phenomenon. Cross-sectional momentum *inverts*. Relative to the market, a stock's intraday move tends to reverse. A likely reason is that the price-contingent flows of §2.5 are market-level flows: index options, index leveraged ETFs and index futures hedges. The flows that reach individual stocks within the day are mostly liquidity demands that get absorbed and laid off. The pattern is **[Fact]**. The reason is my **[Hypothesis]**.

Evidence. Baltussen, Da, Lammers & Martens (2021) for the index. Herberger, Horn & Oehler (2020) and Heston, Korajczyk & Sadka (2010) for the cross-section. §4.4 for where the cross-sectional momentum of the main document is earned.

Verdict. Time-series translates, at the index level. Cross-sectional inverts.

Pitfall. Using an equal-weighted basket of cash stocks as "the market" within the day inherits the staleness of §3.6. Use the futures contract.

5.6 Residual momentum and intraday betas

In the main document. Residual momentum strips out factor exposure before ranking, and has smaller crashes (section 4.6.4 of the main document).

Inside the day. Asynchronous trading, the Epps effect, biases betas estimated from high-frequency returns towards zero. So one-minute residuals still contain much of the market move they were meant to remove. A stock's beta also behaves differently overnight and during the session (Hendershott, Livdan & Rösch, 2020). Residualise against the futures contract, with betas estimated from daily or refresh-time-sampled data, not from one-minute returns.

Evidence. Epps (1979). Budish, Cramton & Shim (2015). No published evidence of profitable *residual* intraday momentum exists.

Verdict. Translates with care, and to no documented effect.

Pitfall. A residual reversal signal built on one-minute residuals is mostly a bet on the part of the market move that the beta missed.

5.7 Oscillators and channel indicators on intraday bars

In the main document. RSI, stochastics, Donchian channels and ADX, with conventional defaults such as fourteen bars (section 4.4 of the main document).

Inside the day. Defaults specified in bars mean something different at each bar size. RSI(14) on one-minute bars covers fourteen minutes, the region where bid–ask bounce and inventory dominate. This makes it a reversal detector for liquidity effects rather than a momentum indicator. The efficiency ratio depends on the bar size even on pure noise (§3.4). Channel rules translate as anchored channels, the ORB (§5.2).

Evidence. Marshall, Cahan & Cahan (2008). Neely & Weller (2003). Schulmeister (2009).

Verdict. Does not translate as momentum. Short-window oscillators on minute bars measure bounce and inventory.

Pitfall. Tuning the defaults on intraday data is risky because the parameter space is large and the effective sample is small (§7.1–7.2).

5.8 Regimes and conditioning

In the main document. Treat a regime as a parameter of the return process, such as its volatility or its trend strength, rather than as a phenomenon (section 1.4 of the main document).

Inside the day. The regime *is the composition of today's flow*. It includes whether dealers are long or short gamma, how large leveraged-ETF assets are relative to volume, and how volatile the morning has been. It also includes whether today holds an option expiry, an index rebalance, a month-end fix or a scheduled announcement. These variables are known, or estimable, before the last half hour. They predict the strength and sign of intraday momentum.

Evidence. Gao, Han, Li & Zhou (2018) on volatility, volume, recessions and news days. Baltussen, Da, Lammers & Martens (2021) on dealers' gamma. Barbon & Buraschi (2020) on the sign flip. Li, Sakkas & Urquhart (2022) on liquidity and information. Rosa (2022) on Markov regimes. Section 10.3 of Dealer Hedging on using the gamma estimate.

Verdict. Translates, and matters more than the signal's functional form.

Pitfall. Regime variables can carry look-ahead. Examples are an end-of-day gamma estimate, the full day's volume, or a news flag stamped with the day's date rather than the time of release.

5.9 Machine-learned intraday signals

In the main document. The main document covers machine learning in a low-signal setting and order-book tensors for intraday prediction (section 4.9 of the main document).

Inside the day. The evidence supports few features. These are time of day, the day's return so far and its components, and the event calendar. They also include the day-level state (overnight gap, opening range, realised volatility) and flow proxies (dealers' gamma, leveraged-ETF assets, published closing imbalances). The labels are the return to the close, or the return over the last half hour. Validation is the main risk. Bars from the same day are strongly dependent through the day's shared state. So cross-validation that shuffles bars leaks information between training and test sets. Purge and embargo by *day*. Section 11.3 of Market Regimes covers the method.

Evidence. The published intraday momentum effects are low-dimensional and conditional. Nothing in §4 suggests that a high-capacity model would find a large effect that a regression misses, outside order-book prediction at horizons of seconds.

Verdict. Translates, with validation by day.

Pitfall. Shuffled cross-validation over bars, and target leakage through overlapping labels such as "return to the close" computed at every bar.

5.10 Calibration: what a good IC is inside the day

In the main document. A monthly cross-sectional IC of 0.02–0.05 is good. A 55% hit rate implies an IC near 0.16, which "should make you suspicious rather than pleased" (section 7.3.4 of the main document).

Inside the day. The best-documented intraday effects have ICs of that size. The rest-of-day effect in equity index futures has $R^2 = 2.45\%$, an IC of $\sqrt{0.0245} \approx 0.157$, and a reported success rate of 55%. By the arcsine identity of the main document's Appendix A.3, $\frac{1}{2} + \arcsin(0.157)/\pi = 55.0\%$. The two numbers agree. Here, though, the high IC does not indicate a problem, for three reasons. The bet is *narrow*: one window a day on one market. It has no large cross-section that would dilute a real edge. The flows behind it are *mechanical*, so no informed trader is being outsmarted. The *capacity* is small: a few million dollars a year of profit at the optimal size in the deepest market (§6.5). So no large trader has a reason to arbitrage the effect away.

The main document's heuristic therefore translates as a different test. **An intraday IC is credible when it comes with a flow mechanism and a small capacity.** A high IC on a broad universe of single stocks, where the mechanism is information and the capacity would be large, deserves the main document's suspicion in full.

Verdict. Inverts in level, translates in spirit.

Pitfall. Taking the level of a published intraday IC as a reason to trust a *different* intraday signal.

5.11 The translation table

Concept in the main document	Verdict inside the day	The one thing to change
Slow information diffusion	Does not apply (seconds to minutes)	Look for flow, not news
Variance ratio VR(q)	Translates with changes	Measure per window and per day, against χ^2_1
Lookback, holding period, the skip	Translates with changes	Anchor windows to the session; skip the bounce
Master form and equivalences	Translates with changes	Weights depend on time of day; many rules are one bet
Volatility normalisation	Translates, with an added role for volatility	Two factors; normalise for risk, condition for strength
Signal transforms	Translates, more important	Dead bands are cost filters
Time-series momentum	Translates at the index level	Trade futures, not baskets
Cross-sectional momentum	Inverts (reversal)	Use it for execution timing only
Residual momentum	Translates with care; no documented effect	Residualise with daily or refresh-time betas
Oscillators and channels	Oscillators do not translate; channels become ORB	Parameters are in clock time, not bars
Regimes	Translates, matters more	The regime is today's flow composition
Machine learning	Translates, with validation by day	Purge and embargo by day, not by bar
IC and hit-rate calibration	Inverts in level	Credible if mechanical, narrow and small

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§5 Key takeaways

1. **Rolling windows become anchored ones.** The session is the unit. Lookbacks start at the previous close or the open, holding periods are clock windows, and the overnight return is its own bar.
2. **The master form survives with time-of-day weights.** The equivalences transfer. At a matched window and threshold, the opening-range breakout, the noise area and the rest-of-day sign rule are close to the same bet. Price minus VWAP is price minus an anchored moving average in volume time.
3. **Normalise for risk, condition for strength.** Inside the day, volatility is both a nuisance and the best predictor of how strong the effect will be.
4. **Dead bands matter more than in the main document**, because they filter costs as well as noise.
5. **Time-series momentum survives at the index level, and cross-sectional momentum inverts.** Residual and oscillator-based versions have no documented intraday edge.
6. **The regime is today's flow composition.** It includes dealers' gamma, leveraged-ETF assets and the event calendar. It matters more than the signal's form.
7. **A high intraday IC is credible when it is mechanical, narrow and small in capacity.** An IC near 0.16 in index futures is plausible. The same IC on a broad universe of single stocks is a bug until proven otherwise.

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6. Implementation: costs, capacity, execution and risk

6.1 What matters, in order

Readers consistently spend their effort in the wrong place. This is the order I would use, most important first.

1. **Timestamps and data.** A one-bar misalignment is worth more than any effect in §4. Get the bar convention, the auction prints and the session boundaries right before anything else (§3.6, §7.3).
2. **The cost and fill model.** At intraday horizons the cost of a trade is of the same order as its expected edge, so a backtest's cost assumptions *are* its result.
3. **Conditioning on the day's state.** Volatility, dealers' gamma, events and the size of the day's move decide whether the effect is present at all (§5.8).

4. **The window.** Use the rest of day rather than the first half hour. Use the last half hour rather than a rolling horizon.
5. **The signal's functional form.** Last, as in the main document.

6.2 The cost wall, derived

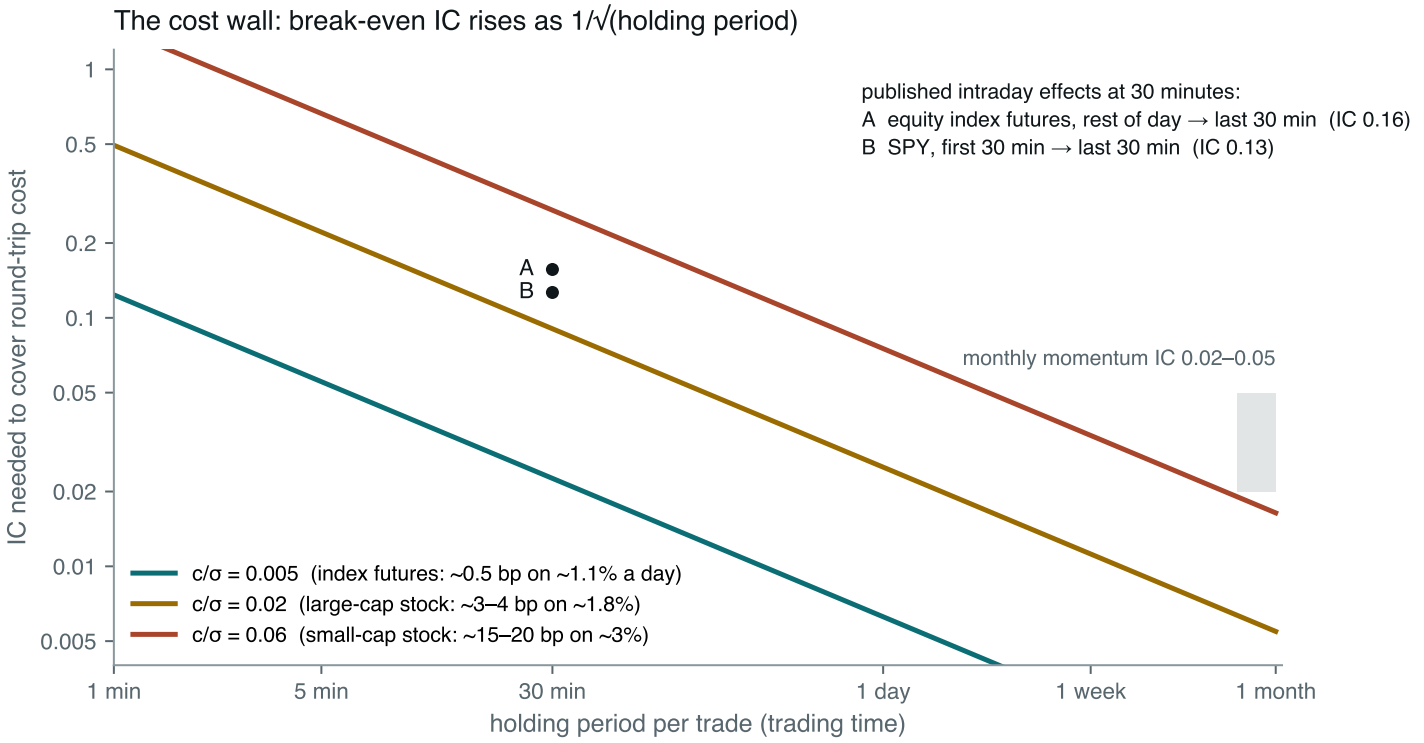
Let a signal s and the return r over the next holding period be jointly normal, with correlation IC and $r \sim N(0, \sigma_h^2)$. A sign strategy holds $\text{sign}(s)$, and its expected return per trade is

$$\mathbb{E}\big[\,\text{sign}(s)\, r\big] \;=\; \mathbb{E}\big[\,\text{sign}(s)\;\mathbb{E}[r \mid s]\big] \;=\; \text{IC}\,\frac{\sigma_h}{\sigma_s}\,\mathbb{E}\,|s| \;=\; \text{IC}\;\sigma_h\sqrt{2/\pi},$$

because $\mathbb{E}[r \mid s] = \text{IC}\,\sigma_h\, s/\sigma_s$ and $\mathbb{E}|s| = \sigma_s\sqrt{2/\pi}$. The trade breaks even when this equals the round-trip cost c:

$$\text{IC}^* \;=\; \frac{c}{\sigma_h}\sqrt{\frac{\pi}{2}} \;\approx\; \frac{c}{\sigma_D}\sqrt{\frac{M}{h}}\,\sqrt{\frac{\pi}{2}},$$

where the second form spreads the daily variance evenly over $M = 390$ one-minute bars (the notation section's bar count, at one-minute granularity). So $\sigma_h \approx \sigma_D\sqrt{h/M}$. That approximation is good to a factor of about 1.5 at any single time of day. It errs in the strategy's favour for the last half hour, which carries more than its even share of variance. Two results follow. The break-even IC scales with the **cost-to-volatility ratio** c/σ_D, a property of the instrument. It also scales as $1/\sqrt{h}$, a property of the horizon. From one month ($h = 21 \times 390$ minutes) to thirty minutes, $\sqrt{8190/30} \approx 16.5$. **At the same cost per trade, a thirty-minute trade needs about sixteen times the IC of a one-month trade.**



The table applies the formula at thirty minutes. The costs are illustrative. They are one tick round trip, or a typical spread for stocks, with nothing for market impact or fees beyond a nominal allowance.

Instrument	Round trip c	Daily volatility	c/σ_D	IC* at 30 min	Documented IC at 30 min
S&P 500 futures (ES)	~0.6 bp	1.1%	0.005	0.025	0.16 (equity index futures, rest of day)

Instrument	Round trip c	Daily volatility	c/σ_D	IC* at 30 min	Documented IC at 30 min
SPY	~0.4 bp	1.1%	0.004	0.016	0.13 (first half hour)
10-year Treasury futures	~1.4 bp	0.45%	0.031	0.14	0.08 (bond futures, pooled)
Crude oil futures	~1.3 bp	2.2%	0.006	0.027	0.04 (commodity futures, pooled)
Euro futures	~0.5 bp	0.5%	0.010	0.045	0.04 (currency futures, pooled)
Large-cap stock	~3–4 bp	1.8%	0.02	0.09	None documented
Small-cap stock	~15–20 bp	3%	0.06	0.26	None documented

Only equity index futures and ETFs clear the wall by a wide margin. Their documented IC is about six times the break-even. Commodity and currency futures sit near break-even. 10-year Treasury futures sit *below* it, because their tick is large relative to their volatility. A bond-futures intraday momentum strategy that crosses the spread every day loses money even if the effect is real. **[Hypothesis]** for these per-instrument readings, since the pooled ICs mix contracts with different costs. The broad conclusion is robust. Intraday momentum is an index-futures phenomenon *economically*, whatever it is statistically.

6.3 Breadth is not free

The fundamental law of active management says the information ratio grows with the square root of the number of independent bets, IR $\approx$ IC$\sqrt{\text{BR}}$. Intraday trading appears to offer many bets. Two things shrink that apparent breadth.

Bets within a day are not independent. They share the day's volatility, the dealers' positioning and the news. With m bets a day whose returns have average pairwise correlation $\bar{\rho}$, the number of *effective* independent bets is

$$m_{\text{eff}} \;=\; \frac{m}{1+(m-1)\,\bar{\rho}},$$

so ten intraday signals with $\bar{\rho} = 0.3$ are worth $10/3.7 \approx 2.7$ independent bets. As m grows, m_{eff} approaches $1/\bar{\rho}$, however many signals are added.

The IC is not constant. For a once-a-day strategy the law predicts $0.126 \times \sqrt{252} \approx 2.0$ for Gao and co-authors' first-half-hour signal, against a realised Sharpe ratio of 1.08. The gap is the variability of the IC itself. An effect concentrated on a few volatile days behaves like a smaller number of larger bets. The kurtosis of 15.65 in the strategy's daily returns reflects the same fact in the P&L.

6.4 Execution: trading the window you forecast

The rest-of-day effect forecasts the move from 3:30 p.m. to the close. Capturing it cleanly takes decisions at both ends.

- **Entry at 3:30 p.m.** Liquidity is high late in the session (the right-hand end of the U-shaped intraday profile), so the spread and impact are near their lowest. Compute the signal from prices that are final at 3:30:00. The order must reach the market after that. A decision that uses the 3:30:00 print and fills at it is look-ahead.
- **Exit at the close.** A market-on-close order fills at the closing-auction price and avoids paying the spread. You must enter it before the exchange's cutoff, 3:50 p.m. at NYSE and 3:55 p.m. at Nasdaq. So you cannot manage the position after the cutoff. In the final minutes the exchanges publish the closing auction's order imbalance. It directly observes part of the flow that drives the effect. **[Practice]** Desks that can react to it use it.
- **Futures after 4:00 p.m.** S&P 500 futures trade after the cash close, but the effect ends at 4:00 p.m. (Baltussen, Da, Lammers & Martens, 2021). So exit at 4:00 p.m. rather than holding into the futures-only period.
- **Passive entry.** Resting limit orders save the spread, but for a momentum strategy their fills are adversely selected. A buy order rests below the market and fills when sellers push the price down to it. This happens on the days the signal is failing. On the days it works, the price runs away and the order never fills. The spread saved is usually smaller than the edge lost. **[Practice]** This is also a general property of passive fills.

6.5 Capacity

The square-root law of market impact says that trading Q dollars moves the price by about $Y\sigma_D\sqrt{Q/V}$, with Y of order one (section 2.6 of the main document; Tóth, Lempérière, Deremble, de Lataillade, Kockelkoren & Bouchaud, 2011). Here V is the market's daily volume and σ_D is its daily volatility. Fold the constants for entry and exit into Y. A round trip of size Q with expected gross edge Δ (as a return) then makes

$$\pi(Q) \;=\; Q\,\Delta \;-\; Y\sigma_D\,\frac{Q^{3/2}}{\sqrt{V}}.$$

Setting $\pi'(Q) = \Delta - \frac{3}{2}Y\sigma_D\sqrt{Q/V} = 0$ gives the profit-maximising size and the profit at that size:

$$Q^* \;=\; \frac{4}{9}\left(\frac{\Delta}{Y\sigma_D}\right)^2 V, \qquad \pi(Q^*) \;=\; \frac{Q^*\Delta}{3}.$$

At the optimum, impact consumes two-thirds of the gross edge, and the trader keeps one third.

Gao and co-authors' first-half-hour strategy earned about 6.67% a year, or $\Delta \approx 2.6$ basis points a day. With $\sigma_D = 1.1\%$ and $Y = 1$, $Q^*/V = \frac{4}{9}(2.6/110)^2 \approx 2.5 \times 10^{-4}$. SPY trades roughly $30 billion a day, and S&P 500 futures trade a few hundred billion. This puts the optimal position at roughly $7.5 million in SPY and $75 million in ES. The profit at that size in ES is $75\text{m} \times 0.00026/3 \approx \$6{,}500$ a day, or about $1.6 million a year. With $Y = 0.5$, the profit is four times that. These figures are orders of magnitude from a stylised model, not estimates. Conditioning on volatile days raises Δ on the days that matter. The conclusion is robust. **The best-documented intraday momentum effect is worth a few million dollars a year at its best size in the deepest futures market in the world.** That amount is negligible for a large fund. It is also the most convincing explanation of why no one has arbitraged the effect away.

6.6 Risk

- **Be flat at the close.** Everything in this chapter is an intraday bet. Holding a last-half-hour position overnight swaps it for a different bet, on the overnight return. §2.6 shows that this return has its own, very different, dynamics.
- **Put stops where the autocorrelation is positive.** Kaminski & Lo (2014) show that a stop-loss adds expected return when returns are positively autocorrelated and subtracts it when they are not. Section 11.3 of Trend-Following develops the result. Inside the day the surface of §3.3 is negative next to the diagonal. So tight stops, the kind that trigger on a few minutes' noise, sit in the reversal zone and are expected to cost money. Stops should be wide enough to match the scale of the effect's window.
- **Treat event days as a separate regime.** Releases at 8:30 a.m. fall before the open and into the overnight return. Releases at 10:00 a.m. fall into the first hour. Federal Reserve announcements at 2:00 p.m. fall into the middle window. Option expiries, index rebalances (with very large closing imbalances), month-ends and half-day sessions change the flows and the depth. Model them separately or exclude them, but decide before testing.
- **Expect the tails to coincide with the signal.** The effect is strongest on the most volatile days (§4.1). These are also the days of trading halts, widened spreads and gaps. Single-stock limit-up/limit-down pauses and market-wide circuit breakers interrupt trading exactly when a momentum position is largest. In the Flash Crash of May 6, 2010, a large automated sell program in ES traded into disappearing depth. Kirilenko, Kyle, Samadi & Tuzun (2017) find that the most active intraday intermediaries did not change their trading pattern as prices fell. A momentum-following or stop-driven strategy on such a day is on the same side as the cascade and fills at the worst prices of the day.
- **Size by volatility, and cap leverage.** Volatility-targeted sizing stabilises risk (§5.3), but a target set as a share of daily volatility implies leverage on calm days. Cap it, and keep within intraday leverage limits.

6.7 Base rates: who makes money trading within the day

Before building an intraday momentum strategy, look at the populations that already trade within the day.

- **Individual day traders, Taiwan, 1992–2006.** Barber, Lee, Liu & Odean (2014) find that few day traders earn positive abnormal returns net of fees. Skill exists and persists in a thin tail. The 500 top-ranked day traders go on to earn 37.9 basis points a day after fees. The bottom-ranked lose 28.9.
- **Individual day traders, Brazil, 2013–2015 starts.** Chague, De-Losso & Giovannetti (2019) follow everyone who began day trading equity futures and persisted for at least 300 days. Of these traders, 97% lost money. 0.4% earned more than a bank teller's wage of US$54 a day. The best earned US$310 a day, with a standard deviation of US$2,560. The authors find no evidence of learning.

- **High-frequency traders.** Baron, Brogaard, Hagströmer & Kirilenko (2019) find that differences in relative latency account for large differences in high-frequency trading firms' performance. They also find that firms which move up the latency ranking after co-location upgrades trade more profitably.
- **Speed on announcements.** Even at five milliseconds, the profits from trading SPY and ES on macroeconomic surprises are about $19,000 and $50,000 per event (Chordia, Green & Kottimukkalur, 2018).

These results are consistent with the cost wall and the capacity calculation. At retail costs the population loses. The profits that exist are small. They go to the fastest traders or to those with the cheapest execution in the deepest instruments.

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§6 Key takeaways

1. **Get the data and the cost model right before the signal.** A one-bar misalignment or a missing spread is larger than any documented intraday effect.
2. **The break-even IC is $(c/\sigma_h)\sqrt{\pi/2}$.** It scales as $1/\sqrt{h}$. For the same cost, it is about sixteen times higher at thirty minutes than at one month.
3. **Only equity index futures and ETFs clear the cost wall by a wide margin.** Commodity and currency futures sit near break-even. 10-year Treasury futures sit below it. No documented single-stock effect would clear it.
4. **Intraday breadth is overstated.** Bets within a day share its state. An IC concentrated in a few volatile days behaves like fewer, larger bets.
5. **Capacity is small.** At the profit-maximising size, impact consumes two-thirds of the edge. The best-documented effect is worth a few million dollars a year in ES. This is why it survives.
6. **Stops belong where the autocorrelation is positive.** Tight stops sit in the intraday reversal zone and cost money.
7. **The base rates are poor.** 97% of persistent Brazilian futures day traders lost money. In Taiwan, skill exists only in a thin tail.

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7. Testing intraday signals without fooling yourself

Section 7 of the main document covers the general methods: the evaluation ladder, walk-forward validation, the information coefficient, the deflated Sharpe ratio, the Reality Check and the bootstrap. All of them apply. This section covers only what is specific to intraday data. That is a great deal, because intraday data offers more ways to be wrong per unit of truth than any other data.

7.1 The unit of evidence is the day

An intraday dataset has millions of bars, which makes it look very large. For a signal that bets once a day, like the rest-of-day effect, the number of independent observations is the number of *days*. Bars within a day are not independent draws of anything.

For a predictive regression with N days and a small correlation IC, the t-statistic is approximately IC$\sqrt{N}$, so reaching $t = 3$ needs $N \approx 9/R^2$ days:

R^2	Example	Days for $t = 3$	Years
2.5%	Rest of day → last half hour, equity index futures	360	1.4
1.6%	First half hour → last half hour, SPY	560	2.2
0.64%	Rest of day → last half hour, bond futures	1,400	5.6
0.2%	Rest of day → last half hour, commodity and currency futures	4,500	18

Three things make the real requirement larger than the table says.

- **Concentration.** When the effect is concentrated on a few volatile days, the effective number of observations is smaller than N. In SPY, excluding the eighteen months from December 2007 to June 2009 halves the R^2 (§4.1). Always report results with the most volatile 1–5% of days removed, and by year.
- **Clustering in pooled samples.** A regression pooled across 17 equity index futures has 17 times the rows but not 17 times the information, because the same day's shock affects every market. Cluster standard errors by day, and by market when the sample has enough markets, as Baltussen, Da, Lammers & Martens (2021) do. Section 4.4 of Econometrics explains why.
- **Overlap.** A signal evaluated at every bar with a label such as "return to the close" produces overlapping labels within each day. Treat the day as the cluster, or keep one observation per day.

7.2 The multiple-testing space is larger than it looks

Section 7.6 of the main document gives the expected best-of-N Sharpe ratio under the null, $\mathrm{SE}(\mathrm{SR})\sqrt{2\ln N}$, about 1.2 for a thousand trials on ten years. Intraday research generates trials faster than any other kind, because researchers can divide the day in so many ways.

- **Windows.** A window is a contiguous run of bars. Thirteen half hours plus the overnight bar give $14 \times 15/2 = 105$ windows to use as predictors, and as many as targets. Five-minute bars give $79 \times 80/2 = 3{,}160$ of each, and millions of predictor–target pairs.
- **Everything else.** Thresholds, stops, exits, bar sizes, sizing rules, event filters and instruments multiply that count again.

The trials are correlated, so the effective number is far smaller than the raw count. But even 10^4 effective trials on ten years of data give an expected best Sharpe ratio of $0.32 \times \sqrt{2\ln 10^4} \approx 1.4$ under the null, and 10^6 give 1.7.

Applied to the published results, the arithmetic supports one result and weakens another.

- **Baltussen and co-authors' equity result is not selection.** A Sharpe ratio of 1.73 over 45 years has a standard error of about $1/\sqrt{45} = 0.15$. So selection alone would need $\sqrt{2\ln N} \approx 11.6$, or $N \approx 10^{29}$ trials. Something real is there.
- **The noise-area strategy is probably not *pure* selection either.** A Sharpe ratio of 1.33 over seventeen years (standard error 0.24) would need about three million independent trials. But the step from the base version's 0.61 to 1.33 came from a sequence of refinements chosen on the same sample. Apply the deflated Sharpe ratio to that step, not to the base.

Count your trials as you go, including the ones abandoned in a notebook, and deflate by that count (section 7.6.2 of the main document). In intraday work the count is usually larger than you think.

7.3 Look-ahead traps specific to intraday data

The main document's look-ahead catalogue (section 6.10 of the main document) was written for daily bars. Intraday data adds its own.

1. **Bar labels.** Vendors label bars by their start time, their end time, or inconsistently. Using a bar stamped 15:30 that ends at 15:35 as "the 15:30 price" gives five minutes of look-ahead. Verify the convention against a known print before anything else.
2. **Same-bar fills.** A signal computed from a bar's close and filled at that same close assumes zero latency and zero information delay. Fill at the next quote after the decision, plus a latency allowance.
3. **The official close.** The exchange sets the closing-auction price at 4:00 p.m. You cannot make a decision that needs it before the market-on-close cutoff (§6.4).
4. **Universe selection.** Choosing stocks by the whole day's volume, range or news, such as "stocks that gapped and ran", uses information from after the decision. Define every filter as of the decision time, as the relative-volume filter of Zarattini, Barbon & Aziz (2024a) does.
5. **The normaliser.** An intraday volatility profile estimated on the full sample carries information about the future's volatility pattern into the past. Estimate it on trailing data.
6. **The path inside a bar.** Open–high–low–close bars do not say whether the high came before the low. A backtest of a breakout with a stop on such bars must assume an order, and the favourable assumption is look-ahead. Holmberg, Lönnbark & Lundström (2013) develop a test from daily open, high, low and close data precisely because intraday paths were unavailable. With intraday data available, use that data.
7. **Clocks and calendars.** Daylight-saving changes fall on different dates in the US and Europe. This shifts cross-market windows by an hour for several weeks a year. Futures sessions start the previous evening. Half-day sessions close at 1:00 p.m. Each misaligns windows silently.

8. **Event timestamps.** A news or regime flag stamped with the day's date rather than its release time leaks the afternoon into the morning.

7.4 Fill realism

- **Market orders** pay half the spread plus impact on each side. The spread depends on the time of day. It is widest in the first minutes and narrowest in the afternoon.
- **Limit orders** need a queue model. The "touch" assumption, that an order fills whenever the price reaches it, is optimistic. The realistic assumption is that it fills only when the price trades *through* it, or after the volume ahead of it in the queue has traded. Fills are also adversely selected (§6.4).
- **The bounce as edge.** Any rule evaluated on transaction prices at fine resolution "earns" part of the spread (§3.6). If a strategy's P&L collapses when marked at midquotes, it was the bounce.
- **Auctions.** Market-on-close orders fill at the auction price, but with imbalance risk and a cutoff. In an index rebalance, the auction can move against the imbalance side.
- **Latency.** Effects that last for seconds are unavailable to a trader whose decision loop takes seconds. At thirty minutes latency does not matter. At five minutes it can.

7.5 Market-structure breaks

A long intraday sample spans several different markets. US equities alone saw many structural changes. Decimal pricing arrived in 2001. Regulation NMS was adopted in 2005 and in force by 2007. High-frequency trading grew in the second half of the 2000s. The Flash Crash came in 2010, and the limit-up/limit-down regime followed in 2012–2013. The closing auction's share of daily volume rose from 3.1% to 7.5% between 2010 and 2018 (Bogousslavsky & Muravyev, 2023). Daily expiries for S&P 500 options began in 2022. Their same-day hedging redistributes dealers' flows within the day (section 7.9 of Dealer Hedging).

Each break changes a term of the master relation of §1.3: who is in G, how deep the market is in the window, or when the flow executes. A pooled 1974–2020 estimate is an average over regimes that may not include today's. Estimate by sub-period, and weight the recent structure.

7.6 Out-of-sample decay

Every intraday effect with a long enough record shows decay or instability.

- The window in which order imbalances predicted NYSE returns shrank from thirty minutes to five between 1996 and 2002 (Chordia, Roll & Subrahmanyam, 2005).
- The rest-of-day slope fell by a third in equity futures and by half in bond futures between 1974–1999 and 2000–2020. It vanished in currency futures (Baltussen, Da, Lammers & Martens, 2021).
- The first-half-hour effect fails out of sample in several studies (Rosa, 2022; §4.1).
- The overnight drift in ES, was 3.7% a year in its best hour from 1998 to 2020. It has averaged close to zero since 2021, as end-of-day imbalances became less dispersed (Boyarchenko, Larsen & Whelan, 2026).

Across 97 cross-sectional predictors of stock returns, McLean & Pontiff (2016) find returns 26% lower out of sample and 58% lower after publication. No reason exists to expect intraday effects to fare better. Flow-based effects have an additional failure mode: the flow can change. **Monitor the flow, not just the P&L.** If leveraged-ETF assets shrink, dealers' gamma turns positive, or the closing auction reorganises, the effect's premise has changed even though its backtest has not.

7.7 Synthetic nulls for intraday pipelines

The main document recommends running every pipeline on a random walk first. Intraday pipelines need a richer null, because the most dangerous artefacts come from the *data-generating details*: the profile, the bounce, the staleness. The simulation behind the figure in §3.4 is a template. It produced these benchmarks, which a correct pipeline should reproduce.

Null	What a correct pipeline shows	What a broken one shows
Random-walk days with a U-shaped profile, fat tails and persistent volatility	Per-day variance ratios distributed as χ^2_1: 5.0% above 3.84, mean 1.00	A mean away from 1, or a tail that depends on the bar size
The same, run through the noise-area rule	Gross P&L of zero ($t = 0.03$), a hit rate of 50%, entries on 86% of days	Positive P&L, from a leak in the band or the fills

Null	What a correct pipeline shows	What a broken one shows
Midquote random walk plus random bid–ask bounce (5 bp spread)	Lag-1 autocorrelation −0.054 at one minute; a reversal rule earns +0.48 bp per bar on transaction prices and zero at midquotes	Profit that survives marking at midquotes
A cash index built from constituents that open late	Opening correlation of 0.18 between the first five minutes and the next twenty-five in the stale index; −0.02 in the true one	The same “opening momentum” in futures data, which indicates an alignment bug

Two more nulls are cheap and powerful for effects like the rest-of-day predictor. **Shuffle days.** Permute the last half-hour returns across days within volatility buckets. This destroys the link to the rest of the day and keeps the dependence of both on volatility. Re-estimate the slope a thousand times. **Shift the window.** Re-run the analysis with the target moved to a window where no scheduled flow executes, such as 1:00–1:30 p.m. An effect that appears in every window is a bug or a drift term, and not a flow effect.

7.8 A checklist

- ☐ Bar timestamps verified against a known print, with session boundaries, half days and daylight-saving dates handled.
- ☐ Decisions use only prices final at the decision time, with fills at the next quote plus latency.
- ☐ Results marked at midquotes as well as transaction prices.
- ☐ Futures or ETFs used for the index, not the cash index at the open.
- ☐ Volatility profile and every filter estimated on trailing data only.
- ☐ Standard errors clustered by day, and by market in pooled panels.
- ☐ Results shown by year, by volatility tercile, and with the most volatile days removed.
- ☐ Every window, threshold, stop and filter tried is counted, and the Sharpe ratio deflated by the count.
- ☐ Costs at realistic levels for the time of day, plus a sensitivity at twice those levels.
- ☐ The synthetic nulls of §7.7 reproduced.
- ☐ A named flow mechanism, with a way to monitor it after deployment.

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§7 Key takeaways

1. **The unit of evidence is the day.** A once-a-day effect with $R^2 = 0.2\%$ needs about eighteen years to reach $t = 3$. Concentration in volatile days makes the real requirement larger.
2. **Intraday research generates trials faster than any other kind.** Five-minute bars give three thousand windows per side before any threshold or stop. Count the trials and deflate.
3. **The long-sample results are not selection, but the refinements may be.** Baltussen and co-authors’ 45-year Sharpe ratio is far beyond what selection can produce. The noise area’s step from 0.61 to 1.33 is exactly what selection produces.
4. **Intraday look-ahead enters through labels and clocks.** The sources are bar-start timestamps, same-bar fills, the official close, full-day universe filters, full-sample profiles and the order of high and low within a bar.
5. **A long sample spans several markets.** Decimalisation, Regulation NMS, high-frequency trading, the growth of the closing auction and daily option expiries each changed the flows behind the effects.
6. **Every intraday effect with a long record has decayed or become unstable.** Flow effects can vanish when the flow changes, as the overnight drift did after 2020. Monitor the flow, not just the P&L.
7. **Run the intraday nulls.** Profile, bounce and staleness each produce patterns that a naive pipeline will report as alpha.

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8. Practice: who uses intraday momentum, and how

[Practice] Much of this section is inference about how firms work. It draws on public descriptions and the structure of the evidence, not on the firms’ books. Read it as informed inference.

8.1 Who harvests which piece

Participant	Horizon	What they use	Where the edge comes from
High-frequency market makers	Milliseconds to seconds	Order-book state; flow prediction	Speed and queue position (Baron, Brogaard, Hagströmer & Kirilenko, 2019)
High-frequency anticipatory traders	Seconds to minutes	Detecting metaorders from trades and orders	Speed and pattern recognition (Hirschey, 2021; van Kervel & Menkveld, 2019)
Options and delta-one desks	Minutes to the close	Their own hedging needs; estimates of market-wide gamma	Knowledge of positioning (section 9 of Dealer Hedging)
Execution desks and algorithms	Minutes to a day	Intraday profiles, time-of-day periodicity, the closing auction	Lower costs for clients (Heston, Korajczyk & Sadka, 2010)
Futures and systematic managers	Last half hour, and execution	Conditional rest-of-day signals; timing of their own trades	Low costs and many markets (Baltussen, Da, Lammers & Martens, 2021)
Proprietary firms and day traders	Minutes to hours	Breakouts, stocks in play	Attention, discipline and low costs, for the few who have them (§6.7)

The cost wall predicts this pattern. At shorter horizons, the edge is more a matter of speed. At longer horizons, it is more a matter of cost and conditioning. At the middle horizons, the effects of this chapter are most valuable to those who already have to trade.

8.2 Intraday momentum as execution alpha

The most valuable use of this chapter for a slower momentum book is probably better execution of the trades the book already has to make. It is probably not a new strategy. This chapter offers five uses, in rough order of evidence:

1. **Time around the close on large-move days.** When the market has moved strongly by 3:30 p.m., the rest-of-day effect predicts further movement in the same direction into the close. It also predicts a partial reversal over the following one to three days. A buyer on a strong up day therefore does better to buy before the last half hour, or to wait a day, than to buy at the close. A seller on the same day can sell into the move. Baltussen and co-authors draw exactly this conclusion. Investors can benefit by providing liquidity against predictable hedging flows, or by timing trades they had already planned. **[Fact]** for the pattern. **[Hypothesis]** for the size of the saving.
2. **Use the same-time-of-day pattern.** Heston, Korajczyk & Sadka (2010) estimate that timing trades to the half-hour periodicity can save about the effective spread.
3. **Trade momentum names at the close rather than the open.** In US stocks, momentum profits have accrued overnight (§4.4). **[Hypothesis]** A momentum book that rebalances at the next open misses that night's return on its new positions. One that rebalances at the close captures it. For a monthly rebalance this is one night in about twenty-one, so the gain is small. The change adds no cost.
4. **Avoid the open for size.** Spreads are widest and prices noisiest in the first minutes, and the opening auction's price can deviate from the continuous market. Unless the information in the trade decays by the minute, paying the open's costs has little justification.
5. **Be the counterparty to predictable flow.** If late-day hedging flow is predictable in direction, providing the liquidity it needs earns a return. The reversal over the following days is that return.

8.3 What has held up, and what has not

Held up	Weakened	Did not hold
Reversal at short intraday lags	The first half hour as a predictor of the last	Generic technical rules on intraday bars, after costs

Held up	Weakened	Did not hold
Rest of day → last half hour, in equity index futures	Rest of day → last half hour in bonds, commodities and currencies	Cross-sectional intraday momentum
Momentum earned overnight in US stocks	The overnight drift in ES, after 2020	Retail day trading as a population
Same-time-of-day continuation in the cross-section	The window in which order imbalances predict returns	The idea that slow diffusion of public news drives intraday trends
Public news priced in seconds to minutes		

8.4 An honest note on expected performance

- **A single equity index future**, trading the rest-of-day effect in the last half hour, had a gross Sharpe ratio of the order of one historically. It is lower after costs. It is probably lower in the current market structure than in the full-sample estimates (§7.5–7.6). My central expectation for a careful implementation today is a net Sharpe ratio between 0.3 and 0.8 on a single market, before diversification. This is an opinion, not a measurement.
- **Diversified across equity index futures**, the Sharpe ratio is higher. The markets' last half hours are correlated through global risk, so the gain is less than the count of markets suggests (§6.3).
- **Capacity**: millions of dollars of annual profit, not hundreds of millions (§6.5).
- **A backtest above a Sharpe ratio of 2** for an intraday momentum strategy at realistic costs: treat it as a problem with costs, fills or look-ahead. Keep that view until shown otherwise (§7.3–7.4).
- **The reliable payoff** of understanding intraday momentum comes in execution and in risk management. It tells you when a day's move is likely to extend into the close, and when it is likely to reverse.

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§8 Key takeaways

1. **The source of the edge changes along the horizon.** It is speed at the fast end, and cost and conditioning at the slow end.
2. **For a slower momentum book, intraday momentum is execution alpha.** Time trades around the close on large-move days, use the same-time-of-day pattern, avoid the open, and trade momentum names at the close.
3. **What held up.** Short-lag reversal, the rest-of-day effect in equity index futures, the overnight accrual of momentum, and the speed of news.
4. **What did not.** Intraday technical rules after costs, cross-sectional intraday momentum, and retail day trading.
5. **Expect a single-market net Sharpe ratio well below one and capacity in the millions.** Anything much better in a backtest is a problem to find.

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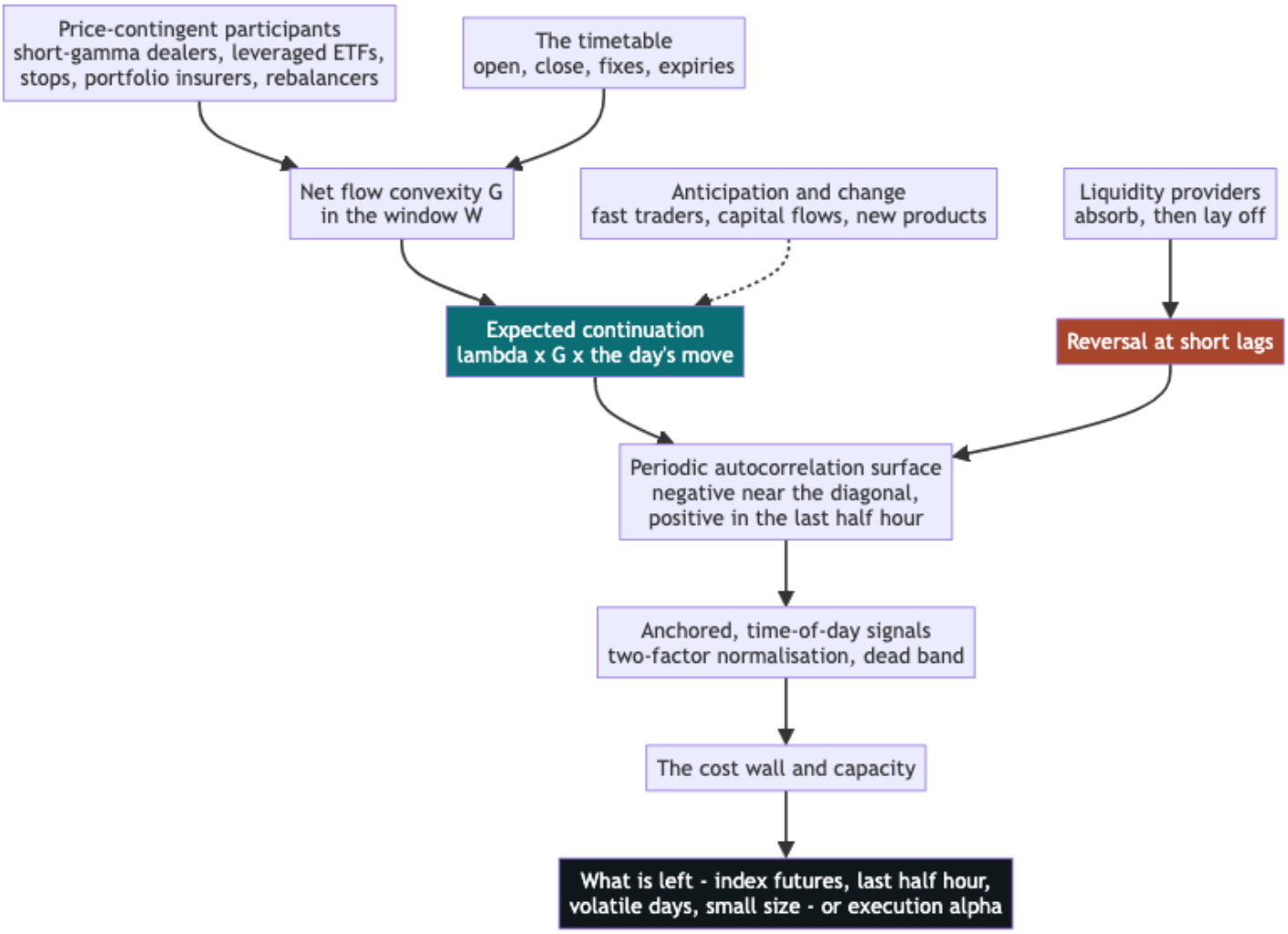
9. Synthesis

9.1 The framework on one page

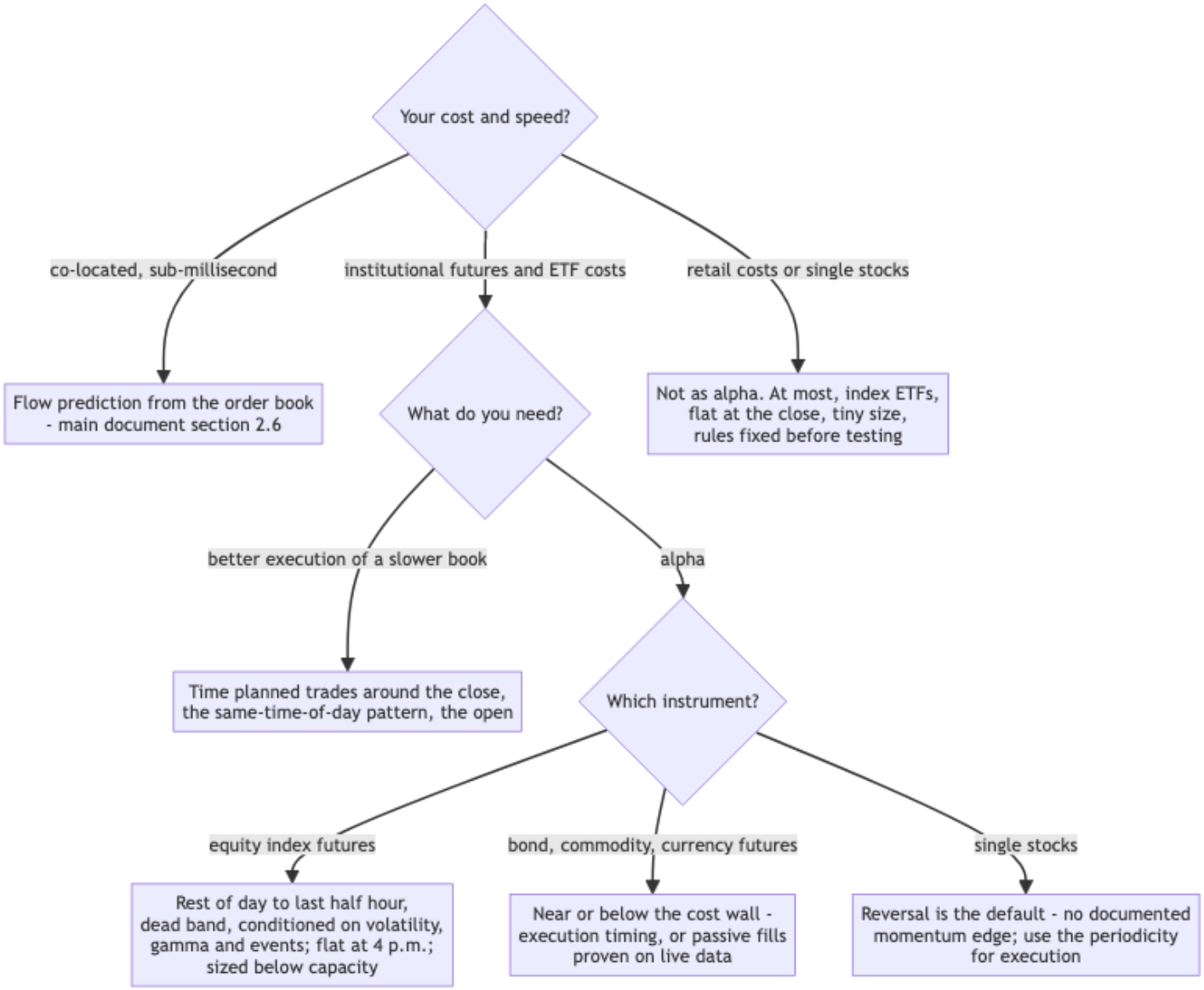
The main document's synthesis has three parts: incomplete adjustment as the mechanism, the variance ratio as the signature, and one master form as the estimator. Inside the day each part has a replacement, and the replacements explain the evidence.

- **Mechanism: incomplete adjustment to flow.** Information is priced in seconds. Inventory and pre-committed, price-contingent flow adjust over hours. Positive-feedback participants create continuation in the window where they execute. Liquidity providers create reversal everywhere else. The master relation $\mathbb{E}[r_{d,W}] \approx \lambda_W \, G \, r_{d,\text{prev close:}\tau}$ has three inputs: who is in G, how deep the market is in the window, and how much of the flow is anticipated. Every result in §4 depends on one of them.
- **Signature: the periodic autocorrelation surface.** It is negative next to the diagonal, positive in the last-half-hour column, and positive at daily lags in the cross-section only. The per-day variance ratio, tested against χ^2_1, is the one-number summary.

- **Estimator: the anchored, time-of-day kernel.** It is normalised by a two-factor volatility, conditioned on the day's state and put through a dead band. The cost wall then judges it. Only equity index futures and ETFs clear the wall by a wide margin.


Price-contingent participants
short-gamma dealers, leveraged ETFs,
stops, portfolio insurers, rebalancers
The timetable
open, close, fixes, expiries
Net flow convexity G
in the window W
Anticipation and change
fast traders, capital flows, new products
Liquidity providers
absorb, then lay off
Expected continuation
lambda x G x the day's move
Reversal at short lags
Periodic autocorrelation surface
negative near the diagonal,
positive in the last half hour
Anchored, time-of-day signals
two-factor normalisation, dead band
The cost wall and capacity
What is left - index futures, last half hour,
volatile days, small size - or execution alpha

9.2 A decision tree



Four steps apply on every branch. Verify timestamps and session boundaries. Mark results at midquotes. Count every window and threshold you try. Name the flow you are betting on, and have a way to watch it.

9.3 A staged build, with gates

Skipping the early stages is the usual cause of failure. These stages are infrastructure work, not research.

Stage	Work	Gate before moving on
0. Data and clocks	Bars with verified timestamps; session calendars; midquotes; auction prints kept separate; futures for indices	The synthetic nulls of §7.7 reproduce
1. Measure the surface	Half-hour periodic autocorrelations for your instruments; the per-day variance-ratio distribution against χ^2_1	The negative band next to the diagonal appears, and the last-half-hour column is measured, not assumed
2. Replicate the published effect	Rest of day → last half hour with the published window definitions; by sub-period; out-of-sample R^2	Sign and rough size replicate in the last ten years, not only over the full sample

Stage	Work	Gate before moving on
3. Condition	Volatility terciles, dealers' gamma, event days, chosen and written down before looking	The effect is monotone in the conditioning variable
4. Costs, execution, capacity	Fills at 3:30 p.m. plus latency; market-on-close exits; costs at twice the estimate; the capacity calculation of §6.5	Positive net Sharpe ratio at twice the costs, at an intended size below half of Q^*
5. Shadow trading	Live signals and simulated fills against real quotes for six to twelve months	Fills and slippage match the model. This stage validates operations, not the effect. A sample of 250 days cannot confirm an R^2 of 2%
6. Small live, monitored	Real size. Track the flow alongside the P&L: leveraged-ETF assets, dealers' gamma and the closing auction's share	A kill rule written before launch, tied to both the P&L and the flow

9.4 Ten things I would tell someone starting today

1. **Start from the flow, not the chart.** Name the participant who must trade, when, in which direction, and why, before testing anything.
2. **Assume reversal until shown otherwise.** It is the default at every short intraday lag.
3. **For alpha, trade index futures or ETFs or nothing.** Everything else is behind the cost wall.
4. **Use the rest of the day, not the first half hour.** The rest of the day is the better predictor. It also survives out of sample.
5. **Condition on the day, and put a dead band on the signal.** The effect appears on volatile days. The dead band covers the cost of the round trip.
6. **Treat the day as the unit of evidence, and the crisis days as the sample.** Show every result with the most volatile days removed.
7. **Count your windows.** Researchers can choose the day's windows in millions of ways, and each choice is a trial.
8. **Mark to midquotes, use futures for the index, and verify every timestamp.** The artefacts of §3.6 are larger than the effects of §4.
9. **Expect small and decaying.** The strategy earns a few million dollars a year at best, and the effect shrinks as the flows change. If you run a slower book, the biggest win is better execution.
10. **Watch the flow composition.** When leveraged-ETF assets, dealers' gamma or the closing auction change, the effect changes. Flow measures show the change before the backtest does.

9.5 What is known, what is not, and what I would bet on

Known. Prices in liquid markets absorb public information within seconds to minutes. Reversal dominates individual securities at intraday lags under an hour. In index futures, the day's move before the last half hour has predicted the last half hour for decades, across four asset classes. Partial reversal follows. In US stocks, the main document's momentum profits have accrued overnight. Generic technical rules on intraday bars do not survive costs and corrections for data snooping. Most people who day trade lose money.

Not known. The evidence does not say how much of the last-half-hour effect comes from hedging, how much from leveraged-ETF rebalancing, how much from scheduled rebalancing, and how much from late-informed trading. It does not say whether daily S&P 500 option expiries have strengthened the effect or spread it over the rest of the day. The evidence on zero-day options' effect on volatility is mixed (section 7.9 of Dealer Hedging). It does not say whether the rest-of-day effect has survived the market structure after 2020. I found no published out-of-sample test of that period. It does not say whether the practitioner breakout systems replicate in independent hands. It does not say what causes the day–night split.

What I would bet on. The rest-of-day effect in equity index futures survives in a weaker, more conditional form. It is strongest on volatile days when dealers are short gamma. The first-half-hour version will not replicate at its published size. The practitioner systems' headline numbers will not replicate at their published sizes either. For most readers of the main document, the lasting value of intraday momentum is a better understanding of when to trade. It is not a strategy.

The main document's decision tree ends its intraday branch with a warning that, at that horizon, "this is order-flow prediction, not momentum". This chapter finds the warning right at the scale of seconds. There, traders predict the flow from the order book. At the scale of the session, the warning needs a refinement. Traders can predict a narrow slice of the flow from the price itself, because some participants must trade as functions of the price. That slice is momentum in form and

flow in substance. Momentum over months is incomplete adjustment to information. Momentum within a day is incomplete adjustment to obligation.

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10. References

Links go to a free copy where one exists, with the publisher's DOI where useful. Free copies include author pages, SSRN, NBER, arXiv, and central-bank and university repositories. Entries marked *paywalled* have no free copy that I could find. The main document's bibliography already covers Jegadeesh & Titman, Moskowitz, Ooi & Pedersen, the deflated Sharpe ratio and the Reality Check. This chapter does not repeat them unless it relies on them.

10.1 Intraday patterns, clocks and measurement

- **Wood, R. A., McInish, T. H. & Ord, J. K. (1985).** "An Investigation of Transactions Data for NYSE Stocks." *Journal of Finance* 40(3), 723–739. *Paywalled.* — The early documentation of the U shape in intraday returns and volatility (§3.1).
- **Harris, L. (1986).** "A Transaction Data Study of Weekly and Intradaily Patterns in Stock Returns." *Journal of Financial Economics* 16(1), 99–117. *Paywalled.* — Intraday return patterns by time of day and weekday (§3.1).
- **Admati, A. R. & Pfleiderer, P. (1988).** "A Theory of Intraday Patterns: Volume and Price Variability." *Review of Financial Studies* 1(1), 3–40. *Paywalled.* — Why trading concentrates in time. The theory behind the U shape (§3.1).
- **Andersen, T. G. & Bollerslev, T. (1997).** "Intraday Periodicity and Volatility Persistence in Financial Markets." *Journal of Empirical Finance* 4(2–3), 115–158. *Paywalled.* — The multiplicative decomposition of intraday volatility, and why researchers must remove the profile before studying dynamics (§3.1, §3.5).
- **Clark, P. K. (1973).** "A Subordinated Stochastic Process Model with Finite Variance for Speculative Prices." *Econometrica* 41(1), 135–155. *Paywalled.* — Returns in business time. The origin of the clock question (§3.5).
- **Ané, T. & Geman, H. (2000).** "Order Flow, Transaction Clock, and Normality of Asset Returns." *Journal of Finance* 55(5), 2259–2284. *Paywalled.* — In transaction time, returns are close to normal (§3.5).
- **Easley, D., López de Prado, M. & O'Hara, M. (2012).** "Flow Toxicity and Liquidity in a High-Frequency World." *Review of Financial Studies* 25(5), 1457–1493. [DOI] — The volume clock and the VPIN toxicity measure (§3.5).
- **Andersen, T. G. & Bondarenko, O. (2014).** "VPIN and the Flash Crash." *Journal of Financial Markets* 17, 1–46. *Paywalled.* — The critique is that VPIN predicts short-run volatility poorly and peaked after the Flash Crash (§3.5).
- **Roll, R. (1984).** "A Simple Implicit Measure of the Effective Bid-Ask Spread in an Efficient Market." *Journal of Finance* 39(4), 1127–1139. *Paywalled.* — Bid–ask bounce and its autocovariance (§3.6).
- **Lo, A. W. & MacKinlay, A. C. (1990).** "An Econometric Analysis of Nonsynchronous Trading." *Journal of Econometrics* 45(1–2), 181–211. [DOI] — Stale prices create spurious autocorrelation in portfolios and indices (§3.6).
- **Epps, T. W. (1979).** "Comovements in Stock Prices in the Very Short Run." *Journal of the American Statistical Association* 74(366), 291–298. *Paywalled.* — Correlations shrink as the sampling interval falls (§3.6, §5.6).

10.2 How fast prices absorb information

- **Patell, J. M. & Wolfson, M. A. (1984).** "The Intraday Speed of Adjustment of Stock Prices to Earnings and Dividend Announcements." *Journal of Financial Economics* 13(2), 223–252. *Paywalled.* — Trading-rule returns gone within five to ten minutes, even in 1979–1981 (§2.1).
- **Busse, J. A. & Green, T. C. (2002).** "Market Efficiency in Real Time." *Journal of Financial Economics* 65(3), 415–437. [DOI] — Prices absorb CNBC reports within a minute for positive reports and fifteen minutes for negative reports (§2.1).
- **Andersen, T. G., Bollerslev, T., Diebold, F. X. & Vega, C. (2003).** "Micro Effects of Macro Announcements: Real-Time Price Discovery in Foreign Exchange." *American Economic Review* 93(1), 38–62. [DOI] — Announcement surprises produce jumps in exchange rates. Bad news moves prices more (§2.1).
- **Chordia, T., Roll, R. & Subrahmanyam, A. (2005).** "Evidence on the Speed of Convergence to Market Efficiency." *Journal of Financial Economics* 76(2), 271–292. [DOI] — Past returns predict nothing at five minutes. Order imbalances did, for a shrinking window (§1.1, §2.4). It is one of the most important papers for this chapter.
- **Chordia, T., Green, T. C. & Kottimukkalur, B. (2018).** "Rent Seeking by Low-Latency Traders: Evidence from Trading on Macroeconomic Announcements." *Review of Financial Studies* 31(12), 4650–4687. [DOI] — SPY and ES respond within five milliseconds. Profits per event are small (§2.1, §6.7).

10.3 Market intraday momentum

- **Gao, L., Han, Y., Li, S. Z. & Zhou, G. (2018).** "Market Intraday Momentum." *Journal of Financial Economics* 129(2), 394–414. [DOI] — The first half hour predicts the last in SPY, 1993–2013. This paper started the modern literature (§4.1). The chapter quotes figures from the working-paper version.
- **Baltussen, G., Da, Z., Lammers, S. & Martens, M. (2021).** "Hedging Demand and Market Intraday Momentum." *Journal of Financial Economics* 142(1), 377–403. [DOI] — The rest of the day predicts the last half hour in 60+ futures, 1974–2020. The authors link the effect to hedging. It is the best-documented effect (§4.2). If you read one paper from this list, read this one.
- **Li, Z., Sakkas, A. & Urquhart, A. (2022).** "Intraday Time Series Momentum: Global Evidence and Links to Market Characteristics." *Journal of Financial Markets* 57, 100619. [DOI] — Sixteen developed markets. The evidence is strong in sample and thinner out of sample than the paper's summary suggests (§4.1).
- **Rosa, C. (2022).** "Understanding Intraday Momentum Strategies." *Journal of Futures Markets* 42(12), 2218–2234. *Paywalled.* — The critique is that out-of-sample predictability disappears. Thresholds help (§4.1, §5.4).
- **Elaut, G., Frömmel, M. & Lampaert, K. (2018).** "Intraday Momentum in FX Markets: Disentangling Informed Trading from Liquidity Provision." *Journal of Financial Markets* 37, 35–51. [DOI] — Rouble–dollar intraday momentum, attributed to liquidity providers' aversion to overnight positions (§4.1).
- **Zhang, Y., Ma, F. & Zhu, B. (2019).** "Intraday Momentum and Stock Return Predictability: Evidence from China." *Economic Modelling* 76, 319–329. *Paywalled.* — Chinese stocks. The morning predicts the afternoon across the lunch break (§3.3, §4.1).
- **Jin, M., Kearney, F., Li, Y. & Yang, Y. C. (2020).** "Intraday Time-Series Momentum: Evidence from China." *Journal of Futures Markets* 40(4), 632–650. [DOI] — Four Chinese commodity futures (§4.1).
- **Wen, Z., Gong, X., Ma, D. & Xu, Y. (2021).** "Intraday Momentum and Return Predictability: Evidence from the Crude Oil Market." *Economic Modelling* 95, 374–384. *Paywalled.* — The United States Oil Fund, 2006–2018. The overnight component carries the signal (§4.1).
- **Shen, D., Urquhart, A. & Wang, P. (2022).** "Bitcoin Intraday Time Series Momentum." *Financial Review* 57(2), 319–344. [DOI] — A market with no close, using volume to define the session (§4.1).
- **Wen, Z., Bouri, E., Xu, Y. & Zhao, Y. (2022).** "Intraday Return Predictability in the Cryptocurrency Markets: Momentum, Reversal, or Both." *North American Journal of Economics and Finance* 62, 101733. [DOI] — Both momentum and reversal across cryptocurrencies (§4.1).
- **Renault, T. (2017).** "Intraday Online Investor Sentiment and Return Patterns in the U.S. Stock Market." *Journal of Banking & Finance* 84, 25–40. [DOI] — First half-hour change in StockTwits sentiment predicts SPY's last half hour (§2.2).

10.4 The cross-section and the day–night split

- **Heston, S. L., Korajczyk, R. A. & Sadka, R. (2010).** "Intraday Patterns in the Cross-section of Stock Returns." *Journal of Finance* 65(4), 1369–1407. [DOI] — Same-half-hour continuation for 40+ days. Reversal within the day (§4.3).
- **Murphy, D. P. & Thirumalai, R. S. (2017).** "Short-Term Return Predictability and Repetitive Institutional Net Order Activity." *Journal of Financial Research* 40(4), 455–477. [DOI] — Links the periodicity to repeated institutional trading (§4.3).
- **Lou, D., Polk, C. & Skouras, S. (2019).** "A Tug of War: Overnight versus Intraday Expected Returns." *Journal of Financial Economics* 134(1), 192–213. [DOI] — Momentum profits earned entirely overnight (§2.6, §4.4). Essential for anyone trading the main document's momentum.
- **Bogousslavsky, V. (2021).** "The Cross-Section of Intraday and Overnight Returns." *Journal of Financial Economics* 141(1), 172–194. [DOI] — When in the day each anomaly accrues (§2.6, §4.4).
- **Hendershott, T., Livdan, D. & Rösch, D. (2020).** "Asset Pricing: A Tale of Night and Day." *Journal of Financial Economics* 138(3), 635–662. [DOI] — Beta priced overnight, not intraday (§2.6, §5.6).
- **Akbas, F., Boehmer, E., Jiang, C. & Koch, P. D. (2022).** "Overnight Returns, Daytime Reversals, and Future Stock Returns." *Journal of Financial Economics* 145(3), 850–875. [DOI] — The intensity of the daily opposing flows predicts returns (§2.6).
- **Berkman, H., Koch, P. D., Tuttle, L. & Zhang, Y. J. (2012).** "Paying Attention: Overnight Returns and the Hidden Cost of Buying at the Open." *Journal of Financial and Quantitative Analysis* 47(4), 715–741. [DOI] — Attention buying at the open, reversed during the day (§2.2).
- **Cliff, M., Cooper, M. J. & Gulen, H. (2008).** "Return Differences between Trading and Non-Trading Hours: Like Night and Day." Working paper, SSRN 1004081. — The US equity premium earned overnight (§2.6).
- **Kelly, M. A. & Clark, S. P. (2011).** "Returns in Trading versus Non-Trading Hours: The Difference Is Day and Night." *Journal of Asset Management* 12(2), 132–145. *Paywalled.* — The same finding, independently (§2.6).
- **Boyarchenko, N., Larsen, L. C. & Whelan, P. (2023).** "The Overnight Drift." *Review of Financial Studies* 36(9), 3502–3547. [DOI] — US equity returns concentrated at the European open, linked to end-of-day imbalances (§2.6).

- **Boyarchenko, N., Larsen, L. C. & Whelan, P. (2026).** "The Disappearing Overnight Drift." *Liberty Street Economics*, Federal Reserve Bank of New York, July 1. — The authors' own update. The drift has averaged close to zero since 2021. This is a blog post. The chapter cites it because it is the primary source for the post-2020 data (§2.6, §7.6).

10.5 Flows: hedging, rebalancing, stops, fixes and the close

- **Cheng, M. & Madhavan, A. (2009).** "The Dynamics of Leveraged and Inverse Exchange-Traded Funds." *Journal of Investment Management* 7(4). — The rebalancing arithmetic of leveraged ETFs (§2.5).
- **Ivanov, I. T. & Lenkey, S. L. (2018).** "Do Leveraged ETFs Really Amplify Late-Day Returns and Volatility?" *Journal of Financial Markets* 41, 36–56. [DOI] — Capital flows offset rebalancing. The effect on late-day returns is economically insignificant (§2.5, §4.2).
- **Shum, P., Hejazi, W., Haryanto, E. & Rodier, A. (2016).** "Intraday Share Price Volatility and Leveraged ETF Rebalancing." *Review of Finance* 20(6), 2379–2409. [DOI] — Rebalancing and end-of-day volatility (§2.5).
- **Ni, S. X., Pearson, N. D., Poteshman, A. M. & White, J. (2021).** "Does Option Trading Have a Pervasive Impact on Underlying Stock Prices?" *Review of Financial Studies* 34(4), 1952–1986. [DOI] — Volatility lower when hedgers are net long options, higher when net short (§2.5).
- **Barbon, A. & Buraschi, A. (2020).** "Gamma Fragility." Working paper, University of St. Gallen. — Dealers' gamma imbalance and the sign of intraday autocorrelation (§2.5, §4.2).
- **Bogousslavsky, V. (2016).** "Infrequent Rebalancing, Return Autocorrelation, and Seasonality." *Journal of Finance* 71(6), 2967–3006. [DOI] — The model behind scheduled-rebalancing explanations (§2.5, §4.3).
- **Bogousslavsky, V. & Muravyev, D. (2023).** "Who Trades at the Close? Implications for Price Discovery and Liquidity." *Journal of Financial Markets* 66, 100852. [DOI] — The closing auction's growth, pricing and reversal (§2.5, §3.6).
- **Osler, C. L. (2003).** "Currency Orders and Exchange Rate Dynamics: An Explanation for the Predictive Success of Technical Analysis." *Journal of Finance* 58(5), 1791–1819. [DOI] — Take-profits sit at round numbers and stop-losses just beyond them. This is the microstructure of support and resistance (§2.5, §4.6).
- **Osler, C. L. (2005).** "Stop-Loss Orders and Price Cascades in Currency Markets." *Journal of International Money and Finance* 24(2), 219–241. [DOI] — Price cascades when the price reaches stop clusters (§2.5, §4.6).
- **Melvin, M. & Prins, J. (2015).** "Equity Hedging and Exchange Rates at the London 4 p.m. Fix." *Journal of Financial Markets* 22, 50–72. [DOI] — A price-contingent flow at a scheduled time, with the reversal the master relation predicts (§2.5).
- **Evans, M. D. D. (2018).** "Forex Trading and the WMR Fix." *Journal of Banking & Finance* 87, 233–247. [DOI] — Negative autocorrelation into and out of the fix (§2.3, §4.6).
- **Ito, T. & Yamada, M. (2017).** "Puzzles in the Tokyo Fixing in the Forex Market: Order Imbalances and Bank Pricing." *Journal of International Economics* 109, 214–234. [DOI] — Predictable customer flow at the Tokyo fix (§2.5, §4.6).
- **Patzelt, F. & Bouchaud, J.-P. (2018).** "Universal Scaling and Nonlinearity of Aggregate Price Impact in Financial Markets." *Physical Review E* 97(1), 012304. [DOI] — Aggregate impact is close to linear for small imbalances (§2.5).

10.6 Liquidity provision, metaorders and high-frequency trading

- **Grossman, S. J. & Miller, M. H. (1988).** "Liquidity and Market Structure." *Journal of Finance* 43(3), 617–633. [DOI] — Market makers sell immediacy. Price pressure reverts (§2.3).
- **Jegadeesh, N. (1990).** "Evidence of Predictable Behavior of Security Returns." *Journal of Finance* 45(3), 881–898. *Paywalled.* — Short-horizon reversal (§2.3).
- **Lehmann, B. N. (1990).** "Fads, Martingales, and Market Efficiency." *Quarterly Journal of Economics* 105(1), 1–28. [DOI] — Weekly reversal as liquidity provision (§2.3).
- **Nagel, S. (2012).** "Evaporating Liquidity." *Review of Financial Studies* 25(7), 2005–2039. [DOI] — Reversal returns are liquidity-provision returns, highest when the VIX is high (§2.3).
- **Hendershott, T. & Menkveld, A. J. (2014).** "Price Pressures." *Journal of Financial Economics* 114(3), 405–423. [DOI] — Inventory-driven price pressure of 0.49% with a half-life under a day (§2.3).
- **Herberger, T. A., Horn, M. & Oehler, A. (2020).** "Are Intraday Reversal and Momentum Trading Strategies Feasible? An Analysis for German Blue Chip Stocks." *Financial Markets and Portfolio Management* 34(2), 179–197. [DOI] — No intraday momentum in DAX stocks. Reversal is too small to trade (§2.3, §4.5).
- **van Kervel, V. & Menkveld, A. J. (2019).** "High-Frequency Trading around Large Institutional Orders." *Journal of Finance* 74(3), 1091–1137. [DOI] — High-frequency traders trade against institutional orders and then back-run them (§2.4).
- **Hirschey, N. (2021).** "Do High-Frequency Traders Anticipate Buying and Selling Pressure?" *Management Science* 67(6), 3321–3345. [DOI] — Anticipatory trading ahead of other investors' flow (§2.4).
- **Tóth, B., Lempérière, Y., Deremble, C., de Lataillade, J., Kockelkoren, J. & Bouchaud, J.-P. (2011).** "Anomalous Price Impact and the Critical Nature of Liquidity in Financial Markets." *Physical Review X* 1(2), 021006. — The square-root impact law (§6.5).

- **Budish, E., Cramton, P. & Shim, J. (2015).** "The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response." *Quarterly Journal of Economics* 130(4), 1547–1621. [DOI] — Correlations break down at millisecond horizons. The arms race (§1.4, §3.6).
- **Kirilenko, A., Kyle, A. S., Samadi, M. & Tuzun, T. (2017).** "The Flash Crash: High-Frequency Trading in an Electronic Market." *Journal of Finance* 72(3), 967–998. [DOI] — Intraday intermediation on May 6, 2010 (§6.6).
- **Coval, J. D. & Shumway, T. (2005).** "Do Behavioral Biases Affect Prices?" *Journal of Finance* 60(1), 1–34. [DOI] — Loss-averse traders move afternoon prices. The moves revert within ten minutes (§2.2).
- **Barber, B. M. & Odean, T. (2008).** "All That Glitters: The Effect of Attention and News on the Buying Behavior of Individual and Institutional Investors." *Review of Financial Studies* 21(2), 785–818. [DOI] — Individuals buy attention-grabbing stocks (§2.2).
- **Securities and Exchange Commission (2010).** "Concept Release on Equity Market Structure." Release No. 34-61358. — Names order anticipation and momentum ignition as strategies of concern (§2.4).

10.7 Technical rules and practitioner systems

- **Neely, C. J. & Weller, P. A. (2003).** "Intraday Technical Trading in the Foreign Exchange Market." *Journal of International Money and Finance* 22(2), 223–237. [DOI] — No excess returns under realistic costs and trading hours (§4.7).
- **Marshall, B. R., Cahan, R. H. & Cahan, J. M. (2008).** "Does Intraday Technical Analysis in the U.S. Equity Market Have Value?" *Journal of Empirical Finance* 15(2), 199–210. *Paywalled.* — 7,846 rules on five-minute data. None survive data-snooping corrections (§4.7).
- **Schulmeister, S. (2009).** "Profitability of Technical Stock Trading: Has It Moved from Daily to Intraday Data?" *Review of Financial Economics* 18(4), 190–201. [DOI] — Gross profits on thirty-minute S&P 500 data after daily profits vanished (§4.7).
- **Neely, C. J. & Weller, P. A. (2012).** "Technical Analysis in the Foreign Exchange Market." In *Handbook of Exchange Rates*, Wiley, 343–373. [DOI] — A survey, including the decay of intraday rules (§4.7).
- **Crabel, T. (1990).** *Day Trading with Short Term Price Patterns and Opening Range Breakout.* Traders Press. — The practitioner source of the opening-range breakout. Out of print, with no stable link.
- **Holmberg, U., Lönnbark, C. & Lundström, C. (2013).** "Assessing the Profitability of Intraday Opening Range Breakout Strategies." *Finance Research Letters* 10(1), 27–33. [DOI] — An ORB test on crude oil futures from daily data (§4.8, §7.3).
- **Zarattini, C. & Aziz, A. (2023).** "Can Day Trading Really Be Profitable? Evidence of Sustainable Long-term Profits from Opening Range Breakout (ORB) Day Trading Strategy vs. Benchmark in the US Stock Market." Working paper, SSRN 4416622. — Five-minute ORB on QQQ. The authors are interested parties (§4.8).
- **Zarattini, C., Barbon, A. & Aziz, A. (2024a).** "A Profitable Day Trading Strategy for the U.S. Equity Market." Swiss Finance Institute Research Paper 24-98. — ORB on stocks in play. The authors are interested parties (§4.8).
- **Zarattini, C., Aziz, A. & Barbon, A. (2024b).** "Beat the Market: An Effective Intraday Momentum Strategy for S&P500 ETF (SPY)." Swiss Finance Institute Research Paper 24-97. — The noise area. The authors are interested parties (§4.8).
- **Maróy, Á. (2025).** "Improvements to Intraday Momentum Strategies Using Parameter Optimization and Different Exit Strategies." Working paper, SSRN 5095349. — A follow-up that optimises the noise-area rule. It illustrates how fast the trial count grows (§4.8).

10.8 Who makes money trading within the day

- **Barber, B. M., Lee, Y.-T., Liu, Y.-J. & Odean, T. (2014).** "The Cross-Section of Speculator Skill: Evidence from Day Trading." *Journal of Financial Markets* 18, 1–24. [DOI] — Taiwan day traders: skill in a thin tail (§6.7).
- **Chague, F., De-Losso, R. & Giovannetti, B. (2019).** "Day Trading for a Living?" Working paper, SSRN 3423101. — Brazilian futures day traders: 97% of the persistent lost money (§6.7).
- **Baron, M., Brogaard, J., Hagströmer, B. & Kirilenko, A. (2019).** "Risk and Return in High-Frequency Trading." *Journal of Financial and Quantitative Analysis* 54(3), 993–1024. — Relative latency drives high-frequency performance (§6.7).
- **Kaminski, K. M. & Lo, A. W. (2014).** "When Do Stop-Loss Rules Stop Losses?" *Journal of Financial Markets* 18, 234–254. [DOI] — Stops help under positive autocorrelation and hurt otherwise (§6.6).

10.9 Methodology, decay and critiques

- **McLean, R. D. & Pontiff, J. (2016).** "Does Academic Research Destroy Stock Return Predictability?" *Journal of Finance* 71(1), 5–32. [DOI] — Returns 26% lower out of sample and 58% lower after publication (§7.6).

The main document's section 7 and its bibliography cover the other methods: the deflated Sharpe ratio, the Reality Check, the bootstrap and walk-forward validation.

10.10 Books

- **Harris, L. (2003).** *Trading and Exchanges: Market Microstructure for Practitioners.* Oxford University Press. — The practitioner's microstructure text: order types, auctions, dealers, and why intraday prices move as they do.
- **Hasbrouck, J. (2007).** *Empirical Market Microstructure: The Institutions, Economics, and Econometrics of Securities Trading.* Oxford University Press. — The econometrics of trades and quotes: bounce, price impact, and measuring the information in order flow.
- **Bouchaud, J.-P., Bonart, J., Donier, J. & Gould, M. (2018).** *Trades, Quotes and Prices: Financial Markets Under the Microscope.* Cambridge University Press. — Order-flow long memory, propagators and the square-root law. It is the physics-based complement to Hasbrouck.

10.11 Background works cited in the appendix

- **Glosten, L. R. & Milgrom, P. R. (1985).** "Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders." *Journal of Financial Economics* 14(1), 71–100. [DOI] — Adverse selection and the spread (A.2).
- **Kyle, A. S. (1985).** "Continuous Auctions and Insider Trading." *Econometrica* 53(6), 1315–1335. *Paywalled.* — Linear price impact of order flow (A.5).
- **Lee, C. M. C. & Ready, M. J. (1991).** "Inferring Trade Direction from Intraday Data." *Journal of Finance* 46(2), 733–746. *Paywalled.* — The standard trade-signing algorithm (A.4).
- **Andersen, T. G., Bollerslev, T., Diebold, F. X. & Labys, P. (2003).** "Modeling and Forecasting Realized Volatility." *Econometrica* 71(2), 579–625. — The realised-variance framework (A.14).
- **Corsi, F. (2009).** "A Simple Approximate Long-Memory Model of Realized Volatility." *Journal of Financial Econometrics* 7(2), 174–196. *Paywalled.* — The HAR model (A.18).
- **Gardner, W. A., Napolitano, A. & Paura, L. (2006).** "Cyclostationarity: Half a Century of Research." *Signal Processing* 86(4), 639–697. *Paywalled.* — Survey of periodically correlated processes (A.15).
- **Hurd, H. L. & Miamee, A. (2007).** *Periodically Correlated Random Sequences: Spectral Theory and Practice.* Wiley. — The textbook treatment (A.15).
- **Campbell, J. Y. & Thompson, S. B. (2008).** "Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average?" *Review of Financial Studies* 21(4), 1509–1531. [DOI] — The out-of-sample R^2 (A.19).
- **Clark, T. E. & West, K. D. (2007).** "Approximately Normal Tests for Equal Predictive Accuracy in Nested Models." *Journal of Econometrics* 138(1), 291–311. [DOI] — The nested-forecast test (A.20).
- **Cameron, A. C., Gelbach, J. B. & Miller, D. L. (2011).** "Robust Inference with Multiway Clustering." *Journal of Business & Economic Statistics* 29(2), 238–249. [DOI] — Two-way clustered standard errors (A.21).
- **Grinold, R. C. (1989).** "The Fundamental Law of Active Management." *Journal of Portfolio Management* 15(3), 30–37. *Paywalled.* — IC times the square root of breadth (A.22).
- **Hamilton, J. D. (1989).** "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle." *Econometrica* 57(2), 357–384. *Paywalled.* — The Markov-switching filter (A.24).
- **López de Prado, M. (2018).** *Advances in Financial Machine Learning.* Wiley. — Purged and embargoed cross-validation (A.25).

10.12 If you only read seven things

1. **Baltussen, Da, Lammers & Martens (2021)** — the best-documented intraday momentum effect, its mechanism and its reversal.
2. **Chordia, Roll & Subrahmanyam (2005)** — why price is efficient at five minutes while flow is not.
3. **Gao, Han, Li & Zhou (2018)**, then **Rosa (2022)** — the effect that started the literature, and the critique that it does not hold out of sample.
4. **Lou, Polk & Skouras (2019)** — where the main document's momentum is earned.
5. **Heston, Korajczyk & Sadka (2010)** — the intraday cross-section: reversal within the day, periodicity across days.
6. **Osler (2005)** — stop cascades: the microstructure of breakouts.
7. **Barber, Lee, Liu & Odean (2014)** and **Chague, De-Losso & Giovannetti (2019)** — the base rates, before you start.

Appendix A. Concepts and prerequisites

This appendix explains the concepts that the main text uses without explanation. The intended reader is comfortable with mathematics and has read the main document. This reader has not necessarily seen market microstructure or forecast-evaluation econometrics. The reader may meet "market-on-close order" in §6.4 or "Clark–West test" in §4.1. The reader

then wants the idea and the precise definition, not a citation, and wants to get them without leaving the document.

Entries follow **dependency** order, not alphabetical order: a later entry uses an earlier one. The entries fall into four parts: the mechanics of trading, the participants whose flows the chapter studies, the measurement problems of intraday data, and the statistics of prediction. Cross-references between entries look like "(A.3)". References back to the main text look like "§x.y", as everywhere else. Each entry gives the idea in words first, then the formal definition, then why it appears here, then where to go deeper. The main document already explains the variance ratio, volatility normalisation and the deflated Sharpe ratio. Those concepts get a short entry that points to it.

Index. The table shows where each concept is first used:

Concept	First used	Concept	First used
The order book, the spread and the midquote	§1.1	Bid–ask bounce	§1.4
Market orders, limit orders and adverse selection	§6.4	Nonsynchronous trading and the Epps effect	§3.6
Auctions and market-on-close orders	§2.3	Realised variance and business time	§3.4
Order flow and order imbalance	§1.1	Periodically correlated processes	§3.1
Price impact and the square-root law	§1.3	The chi-squared distribution	§3.4
Speed, and what fast traders do with it	§1.2	VWAP, channels, the opening range and ATR	§4.8
Trading halts	§6.6	The HAR volatility forecast	§5.3
Delta hedging and gamma	§1.3	Predictive regressions and out-of-sample R^2	§3.3
Leveraged and inverse ETFs	§1.3	Tests for nested forecasts	§4.1
Foreign-exchange fixes	§2.3	Clustered standard errors	§4.2
Stop-loss and take-profit orders	§1.3	IC, hit rate and the fundamental law	§1.3
		The best of many backtests	§7.2
		Markov-switching models	§4.1
		Purged cross-validation	§5.9

Part I — The mechanics of trading. The chapter's mechanism is flow. Understanding flow requires three pieces of market mechanics. The first is an order book that quotes two prices. The second is auctions that set the open and the close. The third is a price that moves when someone demands immediate trading.

A.1 The order book, the spread and the midquote

The idea. Most modern markets are continuous auctions that run through an *order book*. The order book is a list of standing offers to buy (bids) and to sell (asks, or offers) at each price. Markets quote two prices, not one. The first is the best price at which you can sell immediately. The second is the best price at which you can buy immediately. The gap between them is the cost of trading immediately.

Formally. With best bid b_t and best ask a_t, the **quoted spread** is $a_t - b_t$. Analysts usually express it relative to the **midquote** $m_t = (a_t + b_t)/2$. A trade at price p has **effective spread** $2\,|p - m_t|$. This is twice the distance from the midquote at the moment of the trade. It captures price improvement and trades that walk the book. The **tick size** is the minimum price increment. It is one cent for US stocks above a dollar and a quarter of an index point for S&P 500 futures. A spread can never be narrower than one tick. A *round trip* buys at the ask and sells at the bid. It pays one full spread in total, half on each leg.

Why it appears here. §6.2 states the cost wall in spreads and ticks. The analyses of §3.6 and §7.4 depend on the difference between transaction prices and midquotes.

Deeper. Harris (2003), *Trading and Exchanges*, parts I–II; Appendix A.8 of the main document.

A.2 Market orders, limit orders and adverse selection

The idea. A **market order** trades immediately at the best available price and pays the spread for that immediacy. A **limit order** names its price and waits. It earns the spread instead of paying it. It may never fill, and its fills are not a random sample. A resting buy order fills when a seller arrives. Sellers are most likely to arrive, and to arrive in size, when the price is about to fall. The limit orders that fill are therefore disproportionately the ones the trader would rather not have filled. This is **adverse selection**.

Formally. In the model of Glosten & Milgrom (1985), a market maker cannot tell informed from uninformed traders. The market maker sets the ask to the expected value of the asset *given that someone wants to buy*. The market maker sets the bid to its expected value given that someone wants to sell. The spread then protects the market maker against informed traders. A passive order's expected fill price is worse than the midquote at the time the trader placed it.

Why it appears here. §6.4 explains why passive entry usually hurts a momentum strategy. §7.4 covers fill realism.

Deeper. Glosten & Milgrom (1985); Harris (2003).

A.3 Auctions and market-on-close orders

The idea. US stock exchanges open and close the day with **call auctions**. Instead of matching orders one at a time, the exchange collects them for a period. It then clears them all at a single price, the one that lets the most shares trade. The official opening and closing prices are these auction prices. Index funds and ETFs use closing prices as their benchmark, so they concentrate their trading in the closing auction.

Formally. A **market-on-close (MOC)** order executes at the closing auction price, which is unknown when the trader enters the order. A *limit-on-close* order adds a price limit. Traders must enter orders before a cutoff: 3:50 p.m. at NYSE and 3:55 p.m. at Nasdaq. In the final minutes the exchanges publish the auction's accumulated **order imbalance**, the excess of buy over sell interest. The closing auction's share of US daily volume rose from 3.1% in 2010 to 7.5% in 2018 (Bogousslavsky & Muravyev, 2023).

Why it appears here. §2.5 treats the close as the point where scheduled flows meet. §3.6 treats auction prints as a measurement artefact. §6.4 covers exiting at the close.

Deeper. Bogousslavsky & Muravyev (2023); Harris (2003).

A.4 Order flow and order imbalance

The idea. Every trade has a buyer and a seller, so "buying pressure" cannot mean that buyers bought more shares than sellers sold. It means that buyers *initiated* more trades. An initiating buyer is willing to pay the ask rather than wait on the bid. **Order flow** is volume signed by who initiated it. **Order imbalance** is its sum over an interval.

Formally. Give trade k of size v_k the sign $q_k = +1$ if it was buyer-initiated and -1 if seller-initiated. Order imbalance over an interval is $\sum_k q_k v_k$, in shares or dollars, sometimes divided by total volume. Public data do not record the initiator, so analysts infer the sign. They classify trades above the midquote as buys and trades below it as sells. For trades exactly at the midquote, they use a *tick test*, the direction of the last price change. This is the procedure of Lee & Ready (1991).

Why it appears here. In §1.1 and §2.4, order imbalances predict returns over a window that shrank from thirty minutes to five. §2.6 covers end-of-day imbalances and the overnight drift.

Deeper. Lee & Ready (1991); Hasbrouck (2007).

A.5 Price impact, Kyle's lambda and the square-root law

The idea. Price impact is how much the price moves per unit of flow. Prices move *because* people trade, not only while they trade. A market maker cannot tell informed from uninformed flow, so the market maker must move the price in the direction of net demand. Demand is evidence about value. Two regularities matter here. For the *net* imbalance of many traders over a window, impact is close to proportional to the imbalance when the imbalance is small. A *single* large order split into many pieces is a **metaorder**. Its impact grows only with the square root of its size.

Formally. In Kyle (1985), the price change is linear in net order flow, $\Delta p = \lambda Q$. Here λ measures illiquidity. Empirically, a metaorder of Q dollars in a market trading V dollars a day with daily volatility σ_D moves the price by about

$$I(Q) \;\approx\; Y \sigma_D \sqrt{Q/V}, \qquad Y \text{ of order } 1,$$

largely independently of how long the execution takes (Tóth, Lempérière, Deremble, de Lataillade, Kockelkoren & Bouchaud, 2011). Aggregate impact is close to linear for small imbalances and concave for large ones (Patzelt & Bouchaud, 2018). After a metaorder ends, part of its impact decays.

Why it appears here. This concept supplies the λ_W of the master relation (§1.3), the order-of-magnitude check in §2.5 and the capacity calculation of §6.5.

Deeper. Section 2.6 and Appendix A.8 of the main document; Bouchaud, Bonart, Donier & Gould (2018).

A.6 Speed, and what fast traders do with it

The idea. At the shortest horizons, the advantage goes to the fastest trader. **Latency** is the time from a market event to a trader's reaction to it. **Co-location** places servers inside the exchange's data centre. Direct data feeds arrive before the consolidated public feed. Co-location and direct feeds reduce latency to microseconds. Strategies that depend on speed include market making and arbitrage between related instruments. Regulators have singled out two more. **Order anticipation** detects a large order in progress and trades ahead of or alongside it. **Momentum ignition** trades aggressively to trigger a rapid price move, for instance through a level where stop-loss orders cluster (A.11). It then trades against the reaction.

Formally. No formula applies. The SEC's 2010 concept release on equity market structure defines the terms.

Why it appears here. §1.4 and §2.4 cover who sees flow first. §6.7 covers who profits. §7.4 covers latency in fill models.

Deeper. Budish, Cramton & Shim (2015); Baron, Brogaard, Hagströmer & Kirilenko (2019); Hirschey (2021).

A.7 Trading halts

The idea. Trading halts are rules that stop trading when prices move too far, too fast. The halt gives liquidity time to return. The rules exist because of days like the Flash Crash of May 6, 2010. They take effect on exactly the days when a momentum position is largest.

Formally. US **market-wide circuit breakers** halt all stock trading for fifteen minutes if, before 3:25 p.m., the S&P 500 falls 7% or 13% from the previous close. They halt trading for the rest of the day at 20%. Single-stock **limit-up/limit-down** rules, phased in from 2013, set price bands around a recent reference price. If a stock's price stays at a band for fifteen seconds, trading in it pauses for five minutes.

Why it appears here. §6.6 covers tail risk. §7.5 covers market-structure breaks.

Deeper. Kirilenko, Kyle, Samadi & Tuzun (2017) on the event that prompted the reforms.

Part II — The participants. The chapter's mechanism is a set of participants who must trade as functions of the price and the clock. This part explains why each must trade.

A.8 Delta hedging and gamma

The idea. An option dealer avoids exposure to the direction of the market. So the dealer hedges each option with a position in the underlying that offsets the option's sensitivity to small price moves. That sensitivity changes as the price moves, so the dealer must adjust the hedge continually. The direction of the adjustment depends on whether the dealer is net long or net short options.

Formally. An option's **delta** is $\partial V/\partial S$, the change in its value per unit change in the underlying's price S. Its **gamma** is $\Gamma = \partial^2 V/\partial S^2$, the change in delta. A dealer with net gamma Γ that keeps delta at zero must trade $-\Gamma\,\Delta S$ shares after a price change ΔS. In dollars, the trade is $-\Gamma S^2 R$ for a return R. A dealer who is **short gamma** ($\Gamma < 0$) therefore buys after rises and sells after falls, which amplifies moves. A dealer who is **long gamma** does the opposite, which damps them.

Why it appears here. In §1.3 and §2.5, short-gamma hedgers are one of the positive-feedback participants. §4.2 and §5.8 treat gamma as the day's regime.

Deeper. Sections 3.4–3.5, 5 and 7 of Dealer Hedging.

A.9 Leveraged and inverse ETFs

The idea. A leveraged or inverse ETF is an exchange-traded fund that promises a fixed multiple ℓ of an index's *daily* return. The multiple is two or three times for leveraged funds and minus one to minus three times for inverse funds. To keep that

promise the fund must reset its exposure every day. §2.5 shows that the reset always trades in the direction of the day's move.

Formally. A fund with net assets A must trade $A\,\ell(\ell-1)R$ at the end of a day on which the index returned R (§2.5). Because the promise is daily, returns over longer periods are not ℓ times the index's returns. They compound, and volatility reduces them.

Why it appears here. In §1.3, §2.5 and §4.2, leveraged-ETF rebalancing is one of the two proposed drivers of the last-half-hour effect.

Deeper. Cheng & Madhavan (2009); Ivanov & Lenkey (2018) for the offsetting effect of investors' capital flows.

A.10 Foreign-exchange fixes

The idea. Currencies trade around the clock with no official close. So benchmark rates are *fixed* at set times: at 4 p.m. in London and around 10 a.m. in Tokyo. Asset managers value portfolios and settle customer trades at these rates. Large volumes of predictable orders therefore execute at the fix, and the dealers who take them must hedge in the minutes around it.

Formally. The London 4 p.m. fix (the WM/Reuters benchmark) derives from transactions and quotes in a short window around 4 p.m. A dealer fills a customer order "at the fix" at that rate. The dealer must then trade the position in the market.

Why it appears here. In §2.3 and §2.5, fixes supply both a scheduled price-contingent flow and a scheduled reversal. §4.6 covers fixes too.

Deeper. Evans (2018); Melvin & Prins (2015); Ito & Yamada (2017).

A.11 Stop-loss and take-profit orders

The idea. These orders wait for a price level and then trade. A **stop-loss** sells after the price has *fallen* through a level, or buys after it has risen through one. It gives positive feedback and adds to the move. A **take-profit** sells when the price *rises* to a level. It gives negative feedback and trades against the move.

Formally. A stop-loss sell at level K becomes a market sell order when the price first trades at or below K. A take-profit sell at K becomes one when the price first trades at or above K. Because stop-losses become market orders, they execute at whatever price the market offers next, which in a fast move can be well beyond K.

Why it appears here. §1.3, §2.5 and §4.6 use these orders, and §4.6 covers Osler's price cascades. §6.6 covers where to place stops.

Deeper. Osler (2003) and Osler (2005); Kaminski & Lo (2014) on stops as a trading rule.

Part III — Measuring intraday returns. Most errors in intraday research occur before the researcher fits any model. This part covers the measurement methods of §3.

A.12 Bid–ask bounce

The idea. If each recorded price is a trade and trades occur at random on the bid or the ask, prices alternate between bid and ask even when value has not changed. This alternation makes consecutive price changes negatively correlated.

Formally. In the model of Roll (1984), the transaction price is $p_t = m_t + \frac{s}{2}\,q_t$. Here m_t is a random walk whose increments have variance σ_b^2 per bar, s is the spread and $q_t = \pm 1$ is independent and equally likely. Then $\Delta p_t = \Delta m_t + \frac{s}{2}(q_t - q_{t-1})$, so

$$\mathrm{Var}(\Delta p_t) = \sigma_b^2 + \frac{s^2}{2}, \qquad \mathrm{Cov}(\Delta p_t, \Delta p_{t-1}) = -\frac{s^2}{4}, \qquad \rho_1 = -\frac{s^2/4}{\sigma_b^2 + s^2/2}.$$

The autocorrelation grows in magnitude as bars shrink, because σ_b^2 shrinks with the bar and s does not. Inverting the covariance gives Roll's estimator of the spread, $s = 2\sqrt{-\mathrm{Cov}}$.

Why it appears here. Bid–ask bounce appears in §3.3, §3.6 and the right panel of the figure in §3.4.

Deeper. Roll (1984); Appendix A.2 of the main document.

A.13 Nonsynchronous trading and the Epps effect

The idea. Different securities trade at different moments, and a price updates only when a trade happens. An index computed from last prices therefore mixes fresh and stale components. It reflects the news gradually, as the stale components trade. This looks like momentum in the index. The same asynchrony makes correlations between assets look weaker at finer sampling. At one-second resolution, two stocks that move together over a day barely seem related.

Formally. Suppose a stock trades in any given interval only with some probability. Its recorded return in an interval is then the sum of its true returns since its last trade. So a portfolio's recorded return is a moving average of true returns, with positive autocorrelation (Lo & MacKinlay, 1990). The **Epps effect** is the fall of measured correlation towards zero as the sampling interval shrinks (Epps, 1979).

Why it appears here. §3.6 covers the stale cash index at the open. §5.6 covers intraday betas.

Deeper. Lo & MacKinlay (1990); Epps (1979).

A.14 Realised variance and business time

The idea. With intraday data, analysts can *measure* a day's variance instead of estimating it from a model. They add up the squared intraday returns. Finer sampling makes the measurement more precise, until bid–ask bounce (A.12) starts adding its own variance. Separately, the rate at which prices move varies in clock time. It follows trading activity. A clock that advances with trades or volume ("business time") makes returns look far more regular.

Formally. **Realised variance** is $\text{RV}_d = \sum_i r_{d,i}^2$. For a continuous price process it converges to the day's integrated variance as the bars shrink. With additive noise of variance ω^2 in each recorded price, $\mathbb{E}[\text{RV}_d] \approx$ integrated variance $+ 2M\omega^2$ for M bars. So the bias grows with the sampling frequency. Five-minute sampling is the usual compromise. A **business-time** clock replaces calendar time t with a count of cumulative trades or volume.

Why it appears here. In §3.4, realised variance is the denominator of the per-day variance ratio. §3.5 covers the choice of clock.

Deeper. Andersen, Bollerslev, Diebold & Labys (2003); Clark (1973); Ané & Geman (2000); Appendix A.4 of the main document.

A.15 Periodically correlated processes

The idea. A seasonal pattern in the mean is familiar. Ice-cream sales peak every summer. A **periodically correlated** process also has seasonality in its *covariances*. How much today's value says about tomorrow's value depends on the season. Intraday returns behave this way, with a period of one day. The correlation between two consecutive five-minute returns differs between 9:35 and 12:35. It is the same at 9:35 on every day.

Formally. A process X_t is periodically correlated with period T if $\mathbb{E}[X_{t+T}] = \mathbb{E}[X_t]$ and $\text{Cov}(X_{t+T}, X_{s+T}) = \text{Cov}(X_t, X_s)$ for all t, s. Stacking each period's T values into a vector turns the process into an ordinary stationary vector process. So the natural estimator is the covariance by time-of-day slot, computed across periods. This is the chapter's $\gamma(i, k)$. Signal processing uses the term *cyclostationary* for the same idea.

Why it appears here. §3.1 and §3.2 use it to turn the main document's autocorrelation function into a surface.

Deeper. Gardner, Napolitano & Paura (2006), a survey; Hurd & Miamee (2007), a textbook.

A.16 The chi-squared distribution

The idea. The chi-squared distribution is the distribution of the square of a standard normal variable. Its mean is one. It is far from symmetric: most draws are small, and a few are large. This shape is why a ratio that averages 1 can still be well above 3 one time in twenty.

Formally. If $Z \sim N(0, 1)$ then $Z^2 \sim \chi_1^2$, with density $f(x) = e^{-x/2}/\sqrt{2\pi x}$ for $x > 0$, mean 1 and variance 2. Its tail probabilities are those of $|Z|$. For example, $P(Z^2 > 2.71) = 10\%$, $P(Z^2 > 3.84) = 5\%$ and $P(Z^2 > 6.63) = 1\%$, because $1.645^2 = 2.71$, $1.96^2 = 3.84$ and $2.576^2 = 6.63$.

Why it appears here. In §3.4, the per-day variance ratio of a random walk is χ_1^2.

Deeper. Any introductory statistics text. The left panel of the figure in §3.4 compares it with simulated days.

A.17 VWAP, channels, the opening range and ATR

The idea. Intraday traders watch three reference levels. The **volume-weighted average price** (VWAP) is the average price at which the session's volume has traded. Institutions use it as the benchmark for judging their execution. A **channel** is the range between the highest and lowest prices over a lookback. The **opening range** is the channel of the first few minutes of the session.

Formally. $\text{VWAP}_{d,i} = \sum_{j \le i} v_j P_{d,j} \big/ \sum_{j \le i} v_j$, where v_j is the volume in bar j and $P_{d,j}$ its price. An N-minute opening-range breakout goes long when the price rises above the highest price of the first N minutes and short when it falls below the lowest. Breakout systems often size their stops with the **average true range** (ATR). The ATR is the average, over the last 14 days, of each day's true range. The true range is the day's high minus its low, widened to include any gap from the previous close.

Why it appears here. The practitioner systems of §4.8 use all four. §5.2 shows that price minus VWAP is an anchored moving-average displacement in volume time.

Deeper. Section 4.4.3 of the main document for channels; Harris (2003) for VWAP as a benchmark.

A.18 The HAR volatility forecast

The idea. The best forecast of tomorrow's volatility mixes recent volatility at several horizons: yesterday's, last week's and last month's. Traders operating at those horizons all contribute to volatility. A regression on those three averages is cheap and hard to beat.

Formally. The heterogeneous autoregressive (HAR) model of Corsi (2009) is $\text{RV}_{d+1} = \beta_0 + \beta_1 \text{RV}_d + \beta_5 \overline{\text{RV}}_{d-4:d} + \beta_{22} \overline{\text{RV}}_{d-21:d} + \varepsilon_{d+1}$, where the bars denote averages of realised variance (A.14) over the past five and twenty-two days. Researchers often fit it to logarithms or square roots of realised variance.

Why it appears here. §5.3 uses it as a candidate for the day-level factor of the two-factor normaliser.

Deeper. Corsi (2009); Appendix A.24 of Dealer Hedging.

Part IV — The statistics of prediction. The evidence of §4 is a set of predictive regressions, tested in and out of sample, often pooled across markets. This part explains the methods well enough for a reader to read those papers critically.

A.19 Predictive regressions and the out-of-sample R^2

The idea. To test whether something known now predicts a later return, regress the later return on it. The in-sample R^2 says how well the regression line explains the data used to fit it. It always overstates the predictor's value. The out-of-sample version re-estimates the regression at each date using only the data available then. It forecasts one step ahead. It compares those forecasts with the simplest alternative, the average return so far.

Formally. The predictive regression is $r_{d+1} = \alpha + \beta x_d + \varepsilon_{d+1}$. Write $\hat{r}_d$ for the forecast from coefficients estimated on data up to the forecast date, and $\bar{r}_d$ for the historical mean up to the same date. The **out-of-sample** R^2 of Campbell & Thompson (2008) is

$$R^2_{\text{OOS}} \;=\; 1 - \frac{\sum_d (r_d - \hat{r}_d)^2}{\sum_d (r_d - \bar{r}_d)^2}.$$

It is positive when the predictive forecasts beat the historical mean and negative when they do worse. A genuine predictor can do worse when its coefficient estimate is noisy (A.20).

Why it appears here. In §3.3, §4.1 and §4.2, the first-half-hour predictor's in-sample R^2 of 1.49% becomes an out-of-sample R^2 of −1.71%. §7.1 also uses it.

Deeper. Campbell & Thompson (2008); section 9.2 of Econometrics.

A.20 Tests for nested forecasts

The idea. Comparing a model with a predictor against a model without it compares *nested* models, and the comparison is biased against the predictor. Even when the predictor genuinely helps, estimating its coefficient adds noise to the forecasts.

This noise can make the bigger model's squared errors larger. Two tests correct for this bias. They explain how a paper can report both a negative out-of-sample R^2 and significant out-of-sample predictability.

Formally. Let $e_{0,d}$ be the error of the historical-mean forecast, $e_{1,d}$ the error of the predictive forecast, and $\hat{r}_{0,d}, \hat{r}_{1,d}$ the two forecasts. The Clark & West (2007) statistic averages

$$f_d \;=\; e_{0,d}^2 - \left[\, e_{1,d}^2 - (\hat{r}_{0,d} - \hat{r}_{1,d})^2 \,\right]$$

and tests whether its mean is positive with a one-sided t-test. The subtracted term removes the noise from estimating the extra coefficient. A **forecast-encompassing** test asks whether an optimal combination of the two forecasts would put any weight on the predictive one. Rejecting "the mean forecast encompasses the predictive forecast" means the predictor adds information.

Why it appears here. In §4.1, Li, Sakkas & Urquhart (2022) report positive out-of-sample R^2 in 5 of 16 markets. They report significant Clark–West statistics in 10 markets and encompassing rejections in 14.

Deeper. Clark & West (2007).

A.21 Clustered standard errors and the effective number of observations

The idea. Ordinary standard errors assume every observation is an independent draw. Observations that share a shock are not independent. Examples are returns from the same day in different markets and bars from the same day. Treating them as independent overstates the precision of every estimate. Clustering computes the standard error while allowing arbitrary correlation *within* each group of related observations.

Formally. For a regression with observations grouped into clusters g, the clustered variance of the coefficients is the sandwich $(X'X)^{-1}(\sum_g X_g'\hat{u}_g\hat{u}_g'X_g)(X'X)^{-1}$. Here X_g and $\hat{u}_g$ are the regressors and residuals of cluster g. Two-way clustering, by day and by market, combines two such terms (Cameron, Gelbach & Miller, 2011). For m observations with common pairwise correlation $\bar{\rho}$, the **effective number** of independent observations is $m/(1+(m-1)\bar{\rho})$.

Why it appears here. §6.3 covers breadth. §7.1 explains why the day is the unit of evidence.

Deeper. Section 4.4 of Econometrics; Cameron, Gelbach & Miller (2011).

A.22 IC, hit rate and the fundamental law

The idea. The information coefficient is the correlation between a forecast and the outcome. Three conversions make it usable. One converts it into the probability of getting the direction right. One converts it into expected profit per trade. One converts it, together with the number of independent bets, into a risk-adjusted return.

Formally. Take a jointly normal forecast and outcome with correlation IC. The hit rate is $\frac{1}{2} + \arcsin(\text{IC})/\pi$, the arcsine identity. A sign strategy earns IC $\sigma_h\sqrt{2/\pi}$ per trade (§6.2). The **fundamental law of active management** (Grinold, 1989) gives an information ratio of about IC$\sqrt{\text{BR}}$ for BR independent bets a year. The law is the arithmetic of adding bets. A set of BR bets, each with expected return proportional to IC and the same standard deviation, has a total expected return proportional to BR $\cdot$ IC. If the bets are independent, their total standard deviation is proportional to $\sqrt{\text{BR}}$. The ratio is IC$\sqrt{\text{BR}}$. The law therefore assumes independent bets and a constant IC. §6.3 shows both assumptions failing inside the day. For a univariate regression, $R^2 = \text{IC}^2$.

Why it appears here. §5.10 covers calibration, §6.2 the cost wall and §6.3 breadth.

Deeper. Sections 7.3.1 and 7.3.4 and Appendix A.3 of the main document.

A.23 The best of many backtests

The idea. The best of many strategies with no edge will look good. How good it looks is predictable. That prediction is the benchmark that any selected backtest must beat.

Formally. For a strategy with no edge, the annualised Sharpe ratio estimated from T years of roughly independent returns has standard error close to $1/\sqrt{T}$. The expected maximum of N independent such estimates is about $\sqrt{2\ln N}/\sqrt{T}$. The $\sqrt{2\ln N}$ comes from the normal tail. The chance that one standard normal draw exceeds x falls roughly like $e^{-x^2/2}$. So the level that one of N draws can be expected to exceed solves $Ne^{-x^2/2} \approx 1$. This gives $x \approx \sqrt{2\ln N}$, about 3.7 for

a thousand draws and 5.3 for a million. The deflated Sharpe ratio turns this into a test that also accounts for the returns' skewness and kurtosis.

Why it appears here. §7.2.

Deeper. Section 7.6 of the main document.

A.24 Markov-switching models

The idea. A Markov-switching model lets the data alternate between a small number of unobserved states, such as calm and turbulent. Each state has its own parameters. The state today makes some states more likely tomorrow than others. The model infers, day by day, the probability of being in each state.

Formally. A hidden state $z_d \in \{1, \dots, K\}$ follows a Markov chain with transition probabilities $P(z_d = k \mid z_{d-1} = j)$. The observations depend on the state. For example, an observation may follow $r_d \mid z_d = k \sim N(\mu_k, \sigma_k^2)$, or a regression may have a slope that depends on k. The likelihood is computed with the filter of Hamilton (1989) and maximised numerically.

Why it appears here. In §4.1, Rosa (2022) uses one to show that intraday predictability varies over time with the strength of the signal. §5.8 also uses it.

Deeper. Hamilton (1989); section 5 of Market Regimes.

A.25 Purged cross-validation

The idea. Cross-validation estimates how a model will do on new data by holding part of the data out. With time series, and especially with intraday data, a held-out observation and a training observation from the same day share information, so the held-out score is too optimistic. The fix is to remove from the training set every observation whose information overlaps the test period, plus a buffer after it.

Formally. For a test fold covering a set of days, **purging** drops every training observation whose label window intersects those days. For a label like "return to the close", purging drops every bar of the same day. An **embargo** also drops training observations from the days immediately after the fold.

Why it appears here. In §5.9, analysts must purge intraday bars by day.

Deeper. López de Prado (2018), *Advances in Financial Machine Learning*, chapter 7; section 11.3 of Market Regimes.

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